

GENERAL REQUIREMENTS

General Comments

Fire is always a concern, whether a building is under construction, is occupied for normal use or is undergoing renovation, restoration, expansion or demolition. But careful planning combined with common sense can make buildings and premises much safer, regardless of the occupancy or other activities at the site.

The primary focus of the requirements in this chapter is making sure the three elements necessary for a fire—ignition source, fuel and oxygen—do not come in contact with one another. NFPA 550 describes in great detail the features of fire safety systems and includes a logic tree called “The Fire Safety Concepts Tree” to graphically show all the possible means of achieving user-defined fire safety objectives. A portion of that tree is reproduced here as Commentary Figure 3 to show how to avoid fire ignition. Activities on this diagram that follow a plus sign (+), also known as an “or” gate, may be undertaken independently of each other to arrive at the desired goal. Alternatives following a dot (•), also known as an “and” gate, must be combined to achieve the desired result.

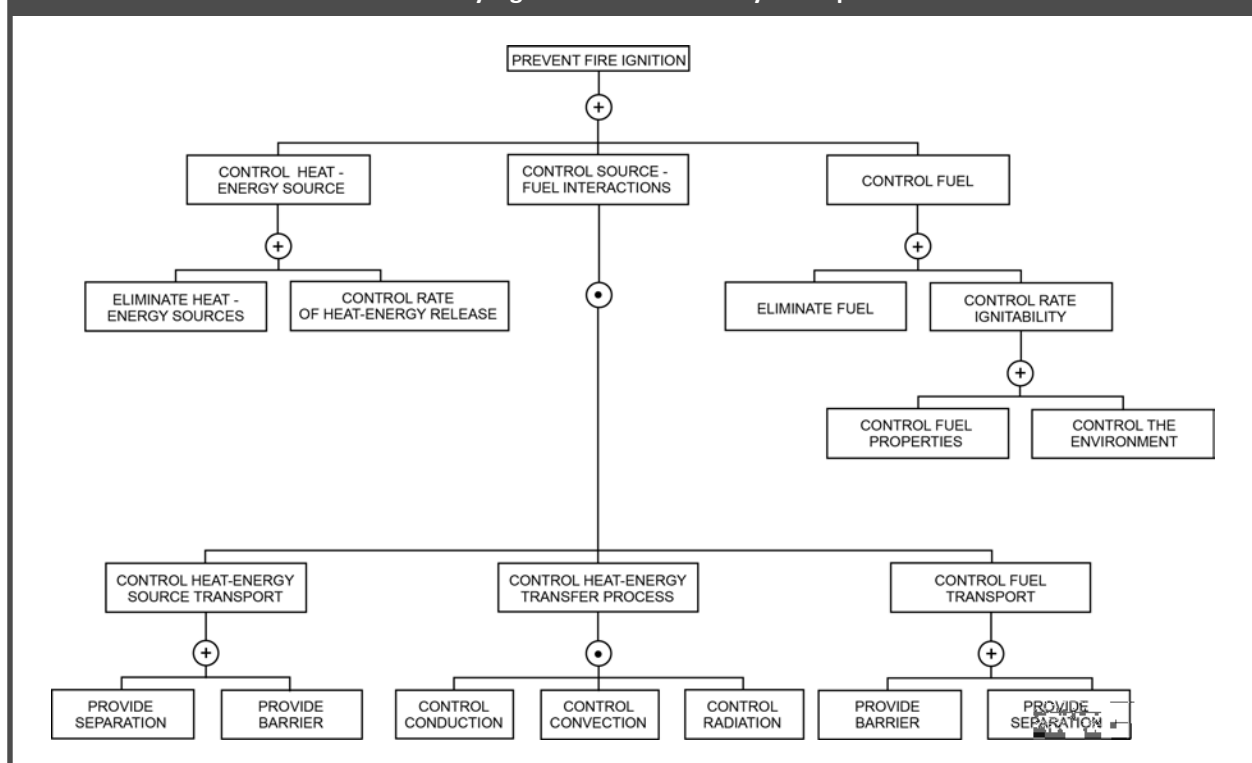
Commentary Figure 3 shows that eliminating any one of the three elements required for a fire to occur will prevent a fire from happening. If there is no ignition source, a fuel load of any size should not catch fire. If there is no fuel load, there is nothing for an ignition source to ignite. If there is little or no air available to sustain combustion, any fire ignited in a fuel load will quickly die.

The requirements and precautions outlined in this chapter, when applied using good judgment and the common sense mentioned above, will help to foster safety for everyone.

Purpose

The requirements and precautions contained in this chapter are intended to improve premises safety for everyone, including construction workers, tenants, operations and maintenance personnel and emergency response personnel.

Commentary Figure 3—The Fire Safety Concepts Tree



GENERAL REQUIREMENTS

SECTION 301—GENERAL

301.1 Scope. The provisions of this chapter shall govern the occupancy and maintenance of all structures and premises for precautions against fire and the spread of fire and general requirements of fire safety.

C The requirements of Chapter 3 prescribe fire safety precautions for conditions that are likely to cause or contribute to the spread of fire in any building or structure or on any premises, regardless of occupancy.

301.2 Permits. Permits shall be required as set forth in Section 105.5 for the activities or uses regulated by Sections 306, 307, 308, 315, 320 and 321.

C Issuing permits gives the fire code official an opportunity to carefully evaluate and regulate hazardous operations. Applicants for permits should be required to demonstrate that their operations comply with the code before the permit is issued. See the commentary to Section 105.5 for a general discussion of operations requiring an operational permit.

SECTION 302—DEFINITIONS

302.1 Definitions. The following terms are defined in Chapter 2:

3D PRINTER.

ADDITIVE MANUFACTURING.

BONFIRE.

HI-BOY.

HIGH-VOLTAGE TRANSMISSION LINE.

OPEN BURNING.

PORTABLE OUTDOOR FIREPLACE.

POWERED INDUSTRIAL TRUCK.

RECREATIONAL FIRE.

SKY LANTERN.

VALET TRASH COLLECTION.

C Definitions of terms can help in the understanding and application of the code requirements. This section directs the code user to Chapter 2 for the proper application of the indicated terms used in this chapter. Terms may be defined in Chapter 2 or in another International Code® (I-Code®) as indicated in Section 201.3, or the dictionary meaning may be all that is needed (see commentary, Sections 201.1 through 201.4).

SECTION 303—ASPALT KETTLES

303.1 Transporting. Asphalt (tar) kettles shall not be transported over any highway, road or street when the heat source for the kettle is operating.

Exception: Asphalt (tar) kettles in the process of patching road surfaces.

C The hazards of hauling a fired kettle of molten asphalt over public ways are obvious. Most asphalt kettles for roofing, paving and similar uses are currently liquefied petroleum gas (LP-gas) fired. Contractors often wish to keep asphalt in a liquid state to save time between jobs and when work is interrupted. Once asphalt is transformed from a solid to a liquid by heating, it retains much of its heat for some time, and although it becomes increasingly viscous as it cools, it remains fluid for a considerable time. Maintaining a fire under a kettle during transport is usually unnecessary and, therefore, prohibited, since little additional heat is required to return the asphalt to a usable consistency. An accident, flat tire or anything else that could cause the kettle to overturn, spilling the molten asphalt in the presence of an open flame, could lead to a serious fire. Even hitting potholes or other bumps in the road could cause the molten asphalt to splash out of the kettle, causing injury to people nearby or damage to property.

The exception for asphalt being used for road repair is necessary for efficient operations for work crews sealing pavement joints and performing similar roadway repairs.

303.2 Location. Asphalt (tar) kettles shall not be located within 20 feet (6096 mm) of any combustible material, combustible building surface or any building opening and within a controlled area identified by the use of traffic cones, barriers or other *approved* means. Asphalt (tar) kettles and pots shall not be utilized inside or on the roof of a building or structure. Roofing kettles and operating asphalt (tar) kettles shall not block *means of egress*, gates, roadways or entrances.

C Asphalt kettles sometimes catch fire. Having one located inside a building would present a serious smoke problem, as well as the fire hazards of asphalt spills flowing to lower floors or the release of LP-gas inside the building. Having one located next to quantities of combustible materials would also represent a fire hazard, as well as the possibility that splashes and splatters could damage construction materials beyond use. Keeping egress pathways and other travel lanes free of

obstructions provides a needed immediate exit from an area where an asphalt kettle-related incident might occur and enhances access to such areas for the fire department.

303.3 Location of fuel containers. Fuel containers shall be located not less than 10 feet (3048 mm) from the burner.

Exception: Containers properly insulated from heat or flame are allowed to be within 2 feet (610 mm) of the burner.

C This section reduces the likelihood that any gas or vapors escaping from fuel containers would be ignited by the open flame of the kettle burner and that the heat of the burner would cause overheating of the fuel containers.

The exception acknowledges the greater safety of insulated containers.

303.4 Attendant. An operating kettle shall be attended by not less than one employee knowledgeable of the operations and hazards. The employee shall be within 100 feet (30 480 mm) of the kettle and have the kettle within sight. Ladders or similar obstacles shall not form a part of the route between the attendant and the kettle.

C Having a trained attendant watch the kettle helps to create a safe operation. The attendant is usually responsible for making sure the asphalt is at the proper temperature, the level of liquid in the kettle is maintained at the required level and the fuel supply for the kettle burner is adequate. The attendant should watch for any change in the kettle that would signal the potential for a safety hazard and adjust the burner output or other factors to keep the kettle in safe operating condition. The attendant is also often responsible for keeping the area surrounding the kettle free of combustible materials and other construction debris that could become a safety hazard.

303.5 Fire extinguishers. There shall be a portable fire extinguisher complying with Section 906 and with a minimum 40-B:C rating within 25 feet (7620 mm) of each asphalt (tar) kettle during the period such kettle is being utilized. Additionally, there shall be one portable fire extinguisher with a minimum 3-A:40-B:C rating on the roof being covered.

C This section defines the type and size of extinguisher that must be available for use, both on the ground near the kettle and on the roof level to which the asphalt is being applied. In the event of a kettle fire, water should not be used as an extinguishing agent because it could cause the molten asphalt to froth and possibly overflow the kettle or spatter over anything or anyone in the surrounding area. See also the commentary to Section 3305.10.2 for roofing operations during construction.

303.6 Lids. Asphalt (tar) kettles shall be equipped with tight-fitting lids.

C A tight-fitting lid on a hot kettle limits the air supply available to feed a kettle fire. Any fire that might start in a closed kettle will quickly burn itself out because of the limited amount of air available for combustion. The lid also helps prevent splashes and splatters that could cause personal injury.

303.7 Hi-boys. Hi-boys shall be constructed of noncombustible materials. Hi-boys shall be limited to a capacity of 55 gallons (208 L). Fuel sources or heating elements shall not be allowed as part of a hi-boy.

C Hi-boys are used on the roof of a building to transport hot asphalt from a point of supply near the edge of the roof to the site of the roofing application. Due to the hazards of molten asphalt discussed in Section 303.1, hi-boys are limited in size to control the maximum amount of potential spills on the roof, which could ignite and pose a high-challenge fire-suppression operation for the fire department. A limited size also enhances the movability and stability of the hi-boy, thus reducing the potential for a tip over. As a further safeguard against a fire incident, hi-boys are prohibited from being fired or equipped with a fuel source for firing. Hi-boys must also be constructed of noncombustible materials to enhance their durability and prevent the container from contributing fuel to a fire. Hi-boys should be well-maintained, including the frame; steering mechanism; tires or wheels; faucets and fill connections (see commentary, Section 202 for the definition of “Hi-boy”).

303.8 Roofing kettles. Roofing kettles shall be constructed of noncombustible materials.

C The requirement for noncombustible materials represents sound safety practice as well as good business practice. Portions of kettles constructed of combustible materials can be easily destroyed and could lead to larger fires. Replacement of destroyed kettles would be expensive. Paying for other fire damage would be even more costly.

Also note that roofing mops soaked in asphalt or pitch must never be left inside a building, near heating equipment or near combustible materials. These mops are subject to spontaneous heating no matter what material they are made of.

303.9 Fuel containers under air pressure. Fuel containers that operate under air pressure shall not exceed 20 gallons (76 L) in capacity and shall be *approved*.

C Limiting the size of pressurized fuel containers limits the probability of a container becoming a major fuel source in case of a kettle fire. Requiring the use of approved containers gives the fire code official more control over the type and suitability of the vessel to be used under pressure.

SECTION 304—COMBUSTIBLE WASTE MATERIAL

304.1 Waste accumulation prohibited. Combustible waste material creating a fire hazard shall not be allowed to accumulate in buildings or structures or upon premises.

C Accumulated waste, trash, construction debris and other natural materials, such as grass clippings, leaves and shrubbery cuttings, can become a serious fire hazard. The three subsections that follow this general statement address the most common situations.

304.1.1 Valet trash. *Valet trash collection* shall be permitted only where *approved*. The *owner* and *valet trash collection* service provider shall comply with the rules and limitations established by the jurisdiction.

C This section makes it clear that, first, the jurisdiction has the authority to allow or prohibit such services and, second, if allowed, it is the owner's responsibility to comply with rules and limitations for the valet trash collection that have been established. Appendix O, if adopted in accordance with Section 101.2.1, provides requirements for valet trash collection in Group R-2 occupancies. These collection services are formally defined in Section 202 as "Valet trash collection," which includes recycling. Occupants receiving this service place trash and recyclables in the corridor outside their residence for pickup by a collection service on a regularly scheduled basis in accordance with restrictions, as prescribed by this appendix.

304.1.2 Waste material. Accumulations of wastepaper, wood, hay, straw, weeds, litter or combustible or flammable waste or rubbish of any type shall not be permitted to remain on a roof or in any *court*, yard, vacant lot, alley, parking lot, open space, or beneath a grandstand, *bleacher*, pier, wharf, manufactured home, recreational vehicle or other similar structure.

C This section considers the kind of waste material that is most likely to accumulate during construction, renovation, additions or demolition and is often referred to as "the housekeeping section." It prohibits disorderly, unkempt storage or accumulation of trash, waste rags, wastepaper, scrub brush and weeds, litter and other combustible materials. Litter and trash represent a serious fire hazard because of their ease of ignition and rapid heat release once ignited. The importance of maintaining property and buildings in good order seems obvious, but sloppy housekeeping still occurs and can be the cause of serious fires. In one of the most serious fires in recent years, improper storage of linseed-oil-soaked rags used to refinish paneling in a high-rise office building caused a fire that destroyed eight floors of the building and killed three firefighters.

304.1.3 Vegetation. Weeds, grass, vines or other growth that is capable of being ignited and endangering property, shall be cut down and removed by the *owner* or occupant of the premises. Vegetation clearance requirements in wildland-urban interface areas shall be in accordance with the *International Wildland-Urban Interface Code*.

C Accumulations of natural waste, such as grass clippings, weed growth and shrubbery cuttings, are not only unsightly, but also represent a serious fire hazard. All too often these accumulations occur at or near fence lines that are adjacent to streets or alleys. This makes accidental ignition by a cigarette butt tossed from a passing vehicle a good possibility. Common sense tells us that removal of this kind of waste is beneficial. The rules of nearly all jurisdictions make waste control and removal the responsibility of the building or property owner, his or her agent, the tenant or the contractor if work is being done on the site. Uncontrolled vegetation growth poses substantial risk to areas designated as wildland-urban interface areas. Accordingly, such areas must comply with the provisions of the *International Wildland-Urban Interface Code*® (IWUIC®).

304.1.4 Space underneath seats. Spaces underneath *grandstand* and *bleacher* seats shall be kept free from combustible and flammable materials. Except where enclosed in not less than 1-hour *fire-resistance-rated* construction in accordance with the *International Building Code*.

C Numerous fires in grandstands and stadiums have shown over the years that the accumulation of flammable or combustible materials under grandstand seating areas can lead to fire disasters. Except as noted in the *International Building Code*® (IBC®), areas under grandstand seating must be kept free of flammable materials, including accumulations of waste or trash. One of the best ways to prevent a fire is to make certain there is no fuel to feed one.

The IBC does allow space under the stands to be used for purposes other than means of egress if that space is separated from the seating area by construction having at least a 1-hour fire-resistance rating. The separation is intended to allow time for occupants in the seating to vacate if a fire should occur. The fire code official would usually have to approve plans for use of space under the stands for concession stands, sales areas or storage areas.

304.1.4.1 Spaces underneath grandstands and bleachers. Spaces underneath *grandstands* and *bleachers* shall not be occupied or utilized for purposes other than *means of egress* except where equipped with an *automatic sprinkler system* in accordance with Section 903.2.1.5.1, or separated with *fire barriers* and *horizontal assemblies* in accordance with Section 1030.1.1.1.

C This section provides specific clarification on the use of spaces under grandstands and bleachers and makes it clear that they are only to be used as a means of egress unless the space is provided with an automatic sprinkler system or is separated by fire barriers and horizontal assemblies. See the commentary to Section 304.1.4 for further information.

304.2 Storage. Storage of combustible rubbish shall not produce conditions that will create a nuisance or a hazard to the public health, safety or welfare.

C Storage of combustible rubbish either indoors or outdoors must be approved by the fire code official. Combustibles should be accumulated in noncombustible containers, such as metal trash cans with tight lids, steel barrels or steel dumpster bins, which should be removed from the site regularly. The use of plastic waste containers should be discouraged due to the extremely high fuel content of such materials, which can sometimes be several times the fuel content of the waste material they contain. Such containers could, under fire conditions, cause rapid fire spread and overtax sprinkler systems, where installed. This section mentions public health as well as safety and welfare, indicating concern over retention of decomposing organic waste as well as flammable and combustible materials.

304.3 Containers. Containers for combustible rubbish and waste material located within or near a structure shall comply with Sections 304.3.1 through 304.3.7.

C Proper containers must be used to improve the safety of indoor or outdoor storage (in close proximity to buildings) of trash and isolate readily combustible materials. This section introduces the more detailed requirements in Sections 304.3.1 through 304.3.7.

304.3.1 Spontaneous ignition. Materials susceptible to spontaneous ignition, such as oily rags, shall be stored in a *listed* disposal container. Contents of such containers shall be removed and disposed of daily.

C Disposal containers, often called “waste cans” or “oily rag cans,” used for storage of materials that might auto-ignite as a result of the spontaneous combustion process must be tested and listed for that use by a recognized testing laboratory or agency and must bear a label showing that they have been tested, along with the name of the testing agency. Such containers are most commonly round and generally available in sizes ranging from 5 to 40 gallons (19 to 151 L). They are equipped with a manual or foot treadle-operated lid that opens to a maximum angle of 60 degrees (1.05 rad) and closes by gravity. These containers are designed to prevent continuing combustion of the contents if ignition occurs. Container design includes features that keep the can body containing waste from coming into contact with combustible surfaces of walls or floors (see commentary, Section 202, for the definition of “Listed”). Daily disposal of container contents reduces the amount of time that oily materials will lie dormant, generating internal heat that can lead to ignition. UL 32 provides further information on the construction, testing and listing of these containers.

304.3.2 Low heat release materials. Where required by this section, low heat release materials shall exhibit a peak rate of heat release not exceeding 300 kW/m² where tested in accordance with ASTM E1354 at an incident heat flux of 50 kW/m² in the horizontal orientation.

C Low heat release materials used in such containers must meet the performance criteria stated when tested in accordance with ASTM E1354. Most nonmetallic waste containers are manufactured from polyethylene, which has a fuel value of 20,050 Btu per pound (46 636 kJ/kg). In comparison, the fuel value of newsprint paper is 9,000 Btu per pound (20 934 kJ/kg). To contain combustible waste in another combustible material that has twice the fuel potential value makes little sense. This section prohibits the use of larger, nonfire-retardant polyethylene trash containers within a structure. Several manufacturers have had the formulation for years to make a fire-retardant polyethylene.

304.3.3 Capacity exceeding 5.33 cubic feet. Containers with a capacity exceeding 5.33 cubic feet (40 gallons) (0.15 m³) shall be provided with lids. Containers and lids shall be constructed of noncombustible materials or low heat release materials in accordance with Section 304.3.2.

C Requiring larger containers to meet stricter conditions is common sense. The larger the volume of waste each container holds, the larger the fire hazard. Isolating the containers from one another with lids helps reduce the possibility that a fire in one container will spread to nearby containers. The lid also helps to smother a fire within the container by limiting the oxygen available to feed it. Additionally, closed containers protect flammable and combustible materials from potential external ignition sources.

304.3.4 Capacity of 1 cubic yard or more. Dumpsters with an individual capacity of 1.0 cubic yard [200 gallons (0.76 m³)] or more shall not be stored in buildings or placed within 5 feet (1524 mm) of combustible walls, openings or combustible roof eave lines unless the dumpsters are constructed of noncombustible materials or low heat release materials in accordance with Section 304.3.2.

Exceptions:

1. Dumpsters in areas protected by an *approved automatic sprinkler system* installed throughout in accordance with Section 903.3.1.1, 903.3.1.2 or 903.3.1.3.
2. Storage in a structure shall not be prohibited where the structure is of Type I or IIA construction, located not less than 10 feet (3048 mm) from other buildings and used exclusively for dumpster or container storage.

C Medium-density polyethylene dumpsters up to 9 cubic yards [27 ft³ (0.76 m³)] in capacity and that have a fuel content of 20,020 Btu per pound (46 567 kJ/kg) [by comparison, municipal solid waste averages 4,500 Btu/lb (10 467 kJ/kg)] are now being distributed and used. Medium-density polyethylene is essentially equal to the fuel value for gasoline and becomes a burning liquid that spreads and flows when involved in a fire. As such, placing them 5 feet (1524 mm) away from a nearby

GENERAL REQUIREMENTS

structure will be of little value unless their flammability hazard is mitigated. This section does that by limiting container heat release rate and heat flux using the test parameters and the peak rate of heat release criterion is provided in Section 304.3.2.

Exception 1 permits storage of these containers indoors if the area is protected by an approved sprinkler system. It would be up to the fire code official to determine the maximum quantities that could be stored under these conditions.

Exception 2 applies only to buildings that are of fire-resistance-rated construction and are used exclusively for waste container storage. Such facilities might be found in scrap yards or at recycling centers, but rarely, if ever, in other occupancies.

304.3.5 Capacity exceeding 1.5 cubic yards. Dumpsters and containers with an individual capacity of 1.5 cubic yards [40.5 cubic feet (1.15 m³)] or more shall not be stored in buildings or placed within 5 feet (1524 mm) of combustible walls, openings or combustible roof eave lines.

Exceptions:

1. Dumpsters or containers that are placed inside buildings in areas protected by an *approved automatic sprinkler system* installed throughout in accordance with Section 903.3.1.1, 903.3.1.2 or 903.3.1.3.
2. Storage in a structure shall not be prohibited where the structure is of Type I or IIA construction, located not less than 10 feet (3048 mm) from other buildings and used exclusively for dumpster or container storage.
3. Dumpsters or containers that are located adjacent to buildings where the exterior area is protected by an *approved automatic sprinkler system*.

C Although waste containers of this size are nearly always constructed of welded steel because of the weight of the waste load, the very fact that the waste load is large makes the containers a large fire hazard. Keeping these large containers in the open and away from combustible construction is the obvious way to keep the fire hazard low.

Exception 1 permits storage of these large containers indoors if the area is protected by an approved sprinkler system. It would be up to the fire code official to determine the maximum quantities that could be stored under these conditions.

Exception 2 applies only to buildings that are of fire-resistance-rated construction and are used exclusively for container storage. Such facilities might be found in scrap yards or at recycling centers, but rarely, if ever, in other occupancies.

Exception 3 permits storage of these large containers less than 5 feet from combustible walls, openings or combustible roof eave lines if the outside area is protected with an automatic sprinkler system.

304.3.6 Waste and linen containers in Group I-1, I-2 and I-3 occupancies and ambulatory care facilities. Waste and linen containers located in Group I-1, I-2 and I-3 occupancies and ambulatory care facilities shall be constructed of noncombustible materials or low heat release materials in accordance with Section 304.3.2. Metal waste containers with a capacity of 20 gallons (75.7 L) or more shall be listed in accordance with UL 1315 and shall be provided with a noncombustible lid. Portable waste and linen containers exceeding 32 gallons (121 L) shall be stored in an area classified as a waste and linen collection room and constructed in accordance with Table 509.1 of the *International Building Code*.

Exception: Recycling clean waste containers complying with Section 304.3.6.2 are not required to be stored in waste and linen collection rooms.

C Although residents of these occupancies are generally closely monitored, there is an increased life safety risk in these facilities since, in each type, the occupants have limited or no self-evacuation ability. Plastic containers can add a tremendous fuel load to a fire in a trash or linen container; many plastic containers will more than triple the fuel load in a fire situation. This section controls the fuel load for these containers, which are used routinely throughout the facilities, and will correlate the code with federal regulations for these facilities. Steps can be taken to reduce the impact of a fire once it has been ignited by limiting wastebaskets to those that meet certain combustibility criteria in accordance with ASTM E1354 as required by Section 304.3.2. ASTM E1354 is a small-scale calorimeter test that measures heat flux output based on a given heat source. This concept is similar to that required in full-scale tests for upholstered furniture and mattresses in Section 805. It should be noted that this section is not focused on ignition prevention but instead assumes that ignition has occurred. The focus is reducing the contribution of the trash container to a fire.

Compliance with this test allows any material that can pass the test to be used. This may allow the use of fire-retardant plastics or other materials that will meet the criteria of the test. Generally, in larger areas of the facility, if a fire can be contained to a wastebasket, it will not spread but will simply burn out.

Since a larger wastebasket means a larger fire, this section also addresses larger metal wastebaskets by requiring compliance with UL 1315 and that they be equipped with a noncombustible lid. This increases the likelihood that the fire will stay contained to the wastebasket and possibly reduce the fire size by restricting the combustion process. The general performance of wastebaskets meeting the requirements of UL 1315 is as follows:

- Limit the external surface temperatures of the container bottom should their contents become ignited.
- Extinguish the fire.
- Contain the contents without contributing fuel to the fire.

See also the commentary to Sections 304.3.2, 304.3.4 and 318.1.

304.3.6.1 Capacity density. The average capacity density of containers located in an individual room or space, other than waste and linen collection rooms, shall not be greater than 0.5 gal/ft (20.4 L/m).

C This section provides a limit on the density of the contents of the containers when they are located in a room or space that is not a waste and linen collection room. The purpose of the limit is to prevent the concentration of these materials so as not to create a hazard that goes beyond the active and passive fire containment systems that are in place within the room or space that the container is located.

304.3.6.2 Recycling clean waste containers. Recycling clean waste containers, including their lids, shall not exceed an individual capacity of 96 gallons (363 L).

C This section provides a limit on the size of individual recycling clean waste containers. Similar to the limit on container density, the purpose is to limit the amount of these recyclables in an individual container so as not to create a hazard that goes beyond the active and passive fire containment systems in place in the room or space that the container is located.

304.3.7 Waste containers with a capacity of 20 gallons or more in Group R-2 college and university dormitories. Waste containers, including their lids, located in Group R-2 college and university dormitories, and with a capacity of 20 gallons (75.7 L) or more, shall be constructed of noncombustible materials or low heat release materials in accordance with Section 304.3.2. Metal waste containers with a capacity of 20 gallons (75.7 L) or more shall be listed in accordance with UL 1315 and shall be provided with a noncombustible lid. Portable waste containers exceeding 32 gallons (121 L) shall be stored in an area classified as a waste collection room constructed in accordance with Table 509.1 of the *International Building Code*.

C This section is similar to Section 304.3.6 but is applicable to larger capacity containers (exceeding 20 gallons) and is focused on Group R-2 college and university dormitories. Large waste containers in college and university dormitories should comply with the same requirements as any waste container in Group I-2 and I-3 occupancies. Most nonmetallic waste containers are manufactured from polyethylene, which has a fuel value more than double that of newsprint and can generate a high heat release rate fire. These waste containers hold combustible waste (much of which is paper) while having very combustible walls. See the commentary to Section 304.3.2 for more information on the test methods prescribed.

SECTION 305—IGNITION SOURCES

305.1 Clearance from ignition sources. Clearance between ignition sources, such as luminaires, heaters, flame-producing devices and combustible materials, shall be maintained in an *approved* manner.

C Establishing safe clearances will usually mean following the requirements of the IBC or other codes adopted by the jurisdiction as well as having the approval of the fire code official.

305.2 Hot ashes and spontaneous ignition sources. Hot ashes, cinders, smoldering coals or greasy or oily materials subject to spontaneous ignition shall not be deposited in a combustible receptacle, within 10 feet (3048 mm) of other combustible material including combustible walls and partitions or within 2 feet (610 mm) of openings to buildings.

Exception: The minimum required separation distance to other combustible materials shall be 2 feet (610 mm) where the material is deposited in a covered, noncombustible receptacle placed on a noncombustible floor, ground surface or stand.

C This section covers two different, but equally serious, ignition source problems. First, hot ashes, embers and cinders from fireplaces, stoves or other fireboxes must never be placed in a combustible container. This point seems almost too obvious to be mentioned, but every year fires are started when someone carelessly scoops ashes containing glowing embers into paper bags or cardboard cartons. It is also not uncommon to see construction scrap being burned in steel drums on construction sites in cold weather. Care must be taken when emptying ashes from those containers to make sure no hot coals get dumped on paper waste or other combustible materials.

The second problem, greasy or oily materials subject to spontaneous combustion, is addressed by requiring such materials to be placed in listed containers (see commentary, Section 304.3.1).

In both cases, safe distances must be maintained from combustible construction and building openings for added protection.

The exception recognizes the added protection of tight-fitting covers on noncombustible trash containers as well as the reduced fire hazard when the containers are placed on a noncombustible surface.

305.3 Open-flame warning devices. Open-flame warning devices shall not be used along an excavation, road, or any place where the dislodgment of such device might permit the device to roll, fall or slide on to any area or land containing combustible material.

C Open-flame warning devices other than fusees used to mark road accidents or other short-term emergencies are rarely used today. The old-fashioned kerosene pots used to mark construction hazards in dark areas have been largely replaced by sawhorse barriers with battery-powered flashing lights. But even though use may be limited, the warning in this section is nonetheless real. One fusee not firmly fixed in the ground or on another stable surface can fall into a roadside ditch filled with dry weeds and cause a roadside fire that could spread into dry woodland or cropland, causing enormous fire damage. Likewise, a burned-out hand-held fusee that is carelessly tossed aside while still hot could ignite dry refuse.

GENERAL REQUIREMENTS

305.4 Deliberate or negligent burning. It shall be unlawful to deliberately or through negligence set fire to or cause the burning of combustible material in such a manner as to endanger the safety of persons or property.

C The deliberate setting of fires, whether in a structure or in a waste container located where it could endanger a structure or its occupants, is normally considered arson, which is a felony that is punishable by a lengthy prison sentence. Fortunately, arson is not that common. More likely, a fire would be caused by carelessness or by someone not considering the consequences of his or her actions.

Regardless of the circumstances, fires must be avoided. Following the requirements in the code as well as those in the IBC will help to maintain a safe, fire-free site. On construction or demolition sites, secure fencing around the site and its waste containers is good protection. Following good housekeeping practices, including routine disposal of combustible materials, is also an excellent first line of protection against fire. An ignition source cannot cause damage to property or endanger life unless there is a fuel load to be ignited.

305.5 Unwanted fire ignitions. Acts or processes that have caused repeated ignition of unwanted fires shall be modified to prevent future ignition.

C Many industrial processes have the potential to produce frequent nuisance fires that generate unwanted alarms necessitating emergency responses that risk the health and safety of firefighters and citizens. This section empowers the fire code official to alter such processes to mitigate nuisance fires.

SECTION 306—MOTION PICTURE PROJECTION ROOMS AND FILM

306.1 Motion picture projection rooms. Electric arc, xenon or other light source projection equipment that develops hazardous gases, dust or radiation and the projection of ribbon-type cellulose nitrate film, regardless of the light source used in projection, shall be operated within a motion picture projection room complying with Section 409 of the *International Building Code*.

C The requirements in this section are specific to spaces housing equipment used to project cellulose acetate film, also called safety film, which is in common use today. This film has about the same fire hazard characteristics as paper of the same thickness and form. The equipment used to project the film, however, may also present fire or health hazards that can be minimized by proper construction of the room. IBC Section 409 covers these construction requirements in detail.

The older type of motion picture film was made of cellulose nitrate, which is also called pyroxylin, which presents a significantly greater fire hazard and, therefore, calls for stricter construction requirements, including sprinklers. These requirements are contained in Sections 903.2.5.3 and 6504. The greater hazard of cellulose nitrate film, which today is found mainly in museum collections and other archives or film preservation facilities, comes from the characteristic of the material to begin degrading at temperatures below its ignition temperature, causing a chemical reaction that can lead to spontaneous combustion. The combustion products of cellulose nitrate are both flammable and extremely toxic because they include oxides of nitrogen. Cellulose nitrate film burns at a rate that is as much as 15 times the rate of common combustibles.

306.2 Cellulose nitrate film storage. Storage of cellulose nitrate film shall be in accordance with NFPA 40.

C NFPA 40 contains minimum requirements for a reasonable level of protection for the storage and handling of cellulose nitrate film. The standard does not address the manufacture of the film because it has not been made in the United States since 1951.

SECTION 307—OPEN BURNING, RECREATIONAL FIRES AND PORTABLE OUTDOOR FIREPLACES

307.1 General. A person shall not kindle or maintain or authorize to be kindled or maintained any *open burning* unless conducted and *approved* in accordance with Sections 307.1.1 through 307.5.

C To control the hazards associated with it, open burning may not be authorized or undertaken without the approvals specified in Section 307. See the commentary to Section 202 for the definition of “Open burning” for a discussion of the types of burning intended to be regulated by this section.

307.1.1 Prohibited open burning. *Open burning* shall be prohibited when atmospheric conditions or local circumstances make such fires hazardous.

Exception: Prescribed burning for the purpose of reducing the impact of wildland fire when authorized by the *fire code official*.

C This section is intended to protect the public from irresponsible burning when it endangers the safety, health or welfare of persons near the burn site. The terms “offensive” and “objectionable” in the previous code edition were undefined and unreasonably subjective, making the prohibition of open burning due to smoke emissions unenforceable. Also, there is not a referenced standard that provides a method for measuring or otherwise determining when smoke emissions are “offensive or objectionable.” In the absence of such guidance, the legal requirements for consistency of enforcement were unattainable.

Fundamentally and historically, the code has regulated fire hazards and not poor air quality caused by smoke from fires. Experience strongly suggests that the previous wording encouraged citizens to request intervention by the fire code official to resolve neighborhood disputes regarding smoke emissions from many forms of outdoor fires that citizens correctly or incorrectly believed to be “open burning” (e.g., recreational fires, bonfires, charcoal burners, etc.); such situations should be civil matters rather than unlawful acts subject to the penalties prescribed in Section 113.

The revised wording empowers the fire code official to prohibit open burning when such fires would, in fact, be hazardous because of measurable environmental factors, such as weather (e.g., wind, temperature, relative humidity) or fuel characteristics (e.g., fuel moisture content).

The exception allows for prescribed burning that is conducted in an effort to minimize the fuel load in wildland and interface areas. Such prescribed burning is a common practice and is done with great planning and forethought and includes safeguards for proper fire control. Annually, several million wildland acres are subjected to prescribed burning in the United States to eliminate hazardous fuels, alter vegetation to promote the growth of fuels that produce less heat and burn slower, improve the habitat for wildlife and vegetation, control nuisance pests, and improve access for replanting and other recreational activities. The ability for a fire agency to utilize prescribed burning operations in a season when the fire can be more easily controlled reduces the personnel and resources needed to control a wildland fire when it occurs and increases the likelihood that fewer structures and lives will be lost to wildland fire.

307.2 Permit required. A permit shall be obtained from the *fire code official* in accordance with Section 105.5 prior to kindling a fire for recognized silvicultural or range or wildlife management practices, prevention or control of disease or pests, or a *bonfire*. Application for such approval shall only be presented by and permits issued to the *owner* of the land on which the fire is to be kindled.

C This section defines a rather narrow range of purposes for which permits will be issued. Section 105.5.36 covers open-burning permits in general. This section restricts permissible fires to those used for silviculture (the cultivation of forests and shade trees); range or wildlife management; pest control; and bonfires as defined in the code. This section further restricts the permitting process to owners of the land on which the fire is to be kindled (see commentary, Section 301.2).

307.2.1 Authorization. Where required by state or local law or regulations, *open burning* shall only be permitted with prior approval from the state or local air and water quality management authority, provided that all conditions specified in the authorization are followed.

C This section requires permit applicants to comply with state and local regulations covering air and water quality as well as safety regulations established by the jurisdiction having authority.

307.3 Extinguishment authority. Where *open burning* creates or adds to a hazardous situation, or a required permit for *open burning* has not been obtained, the *fire code official* is authorized to order the extinguishment of the *open burning* operation.

C Where the fire code official finds that open burning creates a hazardous condition such as the potential for igniting exposures or is unsafe because of atmospheric conditions, this section establishes the authority of the fire code official to order that the open burning fire be extinguished. Note that the fire department is not included in the extinguishment scenario but, rather, the section relies on the provisions of Section 113.3.2, which establishes who must comply with the extinguishment order (i.e., “... the owner, the owner’s authorized agent, operator, occupant or other person responsible for the condition or violation ...”). This enhances public safety by not requiring the commitment of a piece of fire apparatus to extinguish an open-burning fire that is not an emergency (see commentary, Section 113.3.2).

307.4 Location. The location for *open burning* shall be not less than 50 feet (15 240 mm) from any structure, and provisions shall be made to prevent the fire from spreading to within 50 feet (15 240 mm) of any structure.

Exceptions:

1. Fires in *approved* containers that are not less than 15 feet (4572 mm) from a structure.
2. The minimum required distance from a structure shall be 25 feet (7620 mm) where the pile size is 3 feet (914 mm) or less in diameter and 2 feet (610 mm) or less in height.

C The 50-foot (15 240 mm) restriction applies to large fires in large open areas, such as those defined in Section 307.2. Exception 1 refers to fires that generally would be considerably smaller or would be controlled by the container in which they burn, presenting a reduced exposure risk to nearby buildings. Exception 2 allows a reduction in clearance from buildings based on the lesser hazard of fires that are limited in size.

307.4.1 Bonfires. A *bonfire* shall not be conducted within 50 feet (15 240 mm) of a structure or combustible material unless the fire is contained in a barbecue pit. Conditions that could cause a fire to spread within 50 feet (15 240 mm) of a structure shall be eliminated prior to ignition.

C Bonfires usually are large and associated with some kind of planned event (for example, a school pep rally, holiday celebration or camp celebration). This section restricts the location of these large fires to open areas in which sparks and burning embers would be unlikely to endanger structures and smoke would not be a significant hazard to public health. Allowing a bonfire in a barbecue pit automatically restricts the size of the fire to the fuel load that can be contained within the non-combustible fire pit.

GENERAL REQUIREMENTS

307.4.2 Recreational fires. *Recreational fires* shall not be conducted within 25 feet (7620 mm) of a structure or combustible material. Conditions that could cause a fire to spread within 25 feet (7620 mm) of a structure shall be eliminated prior to ignition.

C Recreational fires are usually fairly small but can still represent a significant fire hazard if the area in which they are kindled is not kept free of combustible trash and debris. Basic fire safety practices followed by campers make good guidelines. No fire should ever be kindled in a location where it would endanger structures or would be likely to ignite combustible materials close by.

307.4.3 Portable outdoor fireplaces. Portable outdoor fireplaces shall be used in accordance with the manufacturer's instructions and shall not be operated within 15 feet (4572 mm) of a structure or combustible material.

Exception: Portable outdoor fireplaces used at one- and two-family dwellings.

C Portable outdoor fireplaces (also known as patio fireplaces) designed to burn solid fuel are available at retailers ranging from big box stores to local groceries. Their widespread availability and use created considerable confusion for citizens and the fire service regarding if or how they were regulated by the code since Sections 307 and 308 were essentially silent on the use of this specific type of device. Fires in portable outdoor fireplaces could not be considered a "recreational fire" because critical to that definition is the concept that the fire is not contained in an incinerator, outdoor fireplace, barbecue grill or barbecue pit. It could then have been suggested that a portable outdoor fireplace is merely a type of "outdoor fireplace," but the code did not contain any references pertaining to where an outdoor fireplace could be located or operated. Since the use and hazards associated with operating such fireplaces are more consistent with the type of activities regulated in Section 307 than other types of open flames addressed in Section 308, specific regulations are now provided in Section 307.

This section makes it clear that the use of portable outdoor fireplaces is specifically regulated by the code. It prohibits the use of these devices within 15 feet (3048 mm) of any structure and is consistent with Section 307.4, Exception 1. Also, consistent with regulations for other appliances, this section relies on compliance with the manufacturer's instructions for additional safeguards.

The exception exempts one- and two-family dwellings from the requirements of this section. In those occupancies, the level of familiarity and control exercised by the building occupants is recognized as offsetting the hazards of using these devices. There are practical difficulties involved in enforcing such regulations in one- and two-family dwellings as well.

307.5 Attendance. *Open burning*, bonfires, *recreational fires* and use of portable outdoor fireplaces shall be constantly attended until the fire is extinguished. Not fewer than one portable fire extinguisher complying with Section 906 with a minimum 4-A rating or other *approved* on-site fire-extinguishing equipment, such as dirt, sand, water barrel, garden hose or water truck, shall be available for immediate utilization.

C This section reiterates basic common sense but tends to be ignored quite often. Having one or more individuals responsible for keeping watch on a fire, even one of small size, is the first line of fire prevention. All too often news articles tell of wooden decks burning because hot embers from a charcoal grill fell unobserved onto the unprotected wooden surface, or of a huge brush or forest fire being caused by careless individuals who did not watch their campfires.

For practical purposes as well as for fire safety, some means of extinguishing a kindled fire should be kept close at hand. For small fires, a shovelful of dirt may be sufficient. For large fires, such as bonfires, large volumes of water may be necessary; however, no matter how much extinguishing equipment is available, it may prove useless unless someone is tending the fire and can sound an alarm.

SECTION 308—OPEN FLAMES

308.1 General. Open flame, fire and burning on all premises shall be in accordance with Sections 308.1.1 through 308.4.1 and with other applicable sections of this code.

C This section establishes the scope of the requirements of Section 308 as being applicable to both indoor and outdoor situations involving open flames.

308.1.1 Where prohibited. A person shall not take or utilize an open flame or light in a structure, vessel, boat or other place where highly flammable, combustible or *explosive material* is utilized or stored. Lighting appliances shall be well-secured in a glass globe and wire mesh cage or a similar *approved* device.

C This section is intended to maintain separation between ignitable combustible materials and ignition sources that involve an open flame in any vessel, structure or occupancy.

308.1.2 Throwing or placing sources of ignition. A person shall not throw or place, or cause to be thrown or placed, a lighted match, cigar, cigarette, matches, or other flaming or glowing substance or object on any surface or article where it can cause an unwanted fire.

C This section addresses the hazard caused by carelessness in disposing of smoking materials and other flaming or glowing objects. Smoking in bed or in situations where the smoker could forget about lighted smoking materials has caused large numbers of fires and fatalities over the years. Lack of attention to fireplaces and ash pits has caused great property loss as

well. As one example, a three-story fraternity house at Iowa State University in Ames, Iowa, was completely destroyed by the fire that resulted from a burning log tumbling from an overflowing ash pit onto a combustible floor surface.

Lighted cigarettes discarded through the windows of moving vehicles each year cause grass, brush and forest fires that consume huge acreage in open country as well as dwellings and other structures.

308.1.3 Torches for removing paint. A person utilizing a torch or other flame-producing device for removing paint from a structure shall provide not less than one portable fire extinguisher complying with Section 906 and with a minimum 4-A rating, two portable fire extinguishers, each with a minimum 2-A rating, or a water hose connected to the water supply on the premises where such burning is done. The person doing the burning shall remain on the premises 1 hour after the torch or flame-producing device is utilized.

C Any time an open flame is used to soften old paint in preparation for removal, there is a risk of fire that must be covered by having approved fire extinguishers or a water source readily available. The requirement for a 1-hour fire watch after discontinuing the use of the open flame covers the possibility that paint fragments could still be hot enough to ignite flammable or combustible materials that might be lying around. It also considers the possibility that the flame used to remove paint from a combustible base material could heat that material to its ignition temperature and leave an almost undetectable smolder that might burst into flames later. Safe and effective means for removing paint at lower temperatures, such as warm-air heat devices capable of generating high-temperature convection air, are readily available for sale or rent and far less likely to result in an ignition of combustible materials.

308.1.4 Location near combustibles. Open flames such as from candles and lanterns shall not be located on or near decorative material or similar combustible materials.

C Accidents involving candles and lanterns used in both outdoor and indoor settings are not at all uncommon. In nearly all of these incidents, the exercise of common sense and the practice of keeping ignition sources and fuel packages well separated could have prevented property damage or loss of life.

308.1.5 Open-flame devices in wildfire risk areas. Torches and other devices, machines or processes liable to start or cause fire shall not be operated or used in or on wildfire risk areas, except by a permit in accordance with Section 105.5 secured from the *fire code official*.

Exception: Use within inhabited premises or designated campsites that are not less than 30 feet (9144 mm) from grass-, grain-, brush- or forest-covered areas.

C This section establishes the fire code official's authority to control through the permitting process the use of open flames in wildfire risk areas susceptible to fires. The term "Wildfire risk area" is defined in Chapter 2 and is applied to land covered with grass, grain, brush, forest or similar vegetation that, if ignited, could pose a severe fire danger to surrounding areas. The exception recognizes open ground that is free of combustible materials as an acceptable barrier to fire spread (see the IWUIC for further information).

308.1.5.1 Signals and markers. Flame-employing devices, such as lanterns or kerosene road flares, shall not be operated or used as a signal or marker in or on wildfire risk areas.

Exception: The proper use of fusees at the scenes of emergencies or as required by standard railroad operating procedures.

C This section prohibits the use of flame-producing devices as signal or marker devices except for the use of fusees to mark the scene of an emergency or where routinely employed in railroad procedures, such as when a train is stopped across a roadway not protected by permanent signal lights.

308.1.6 Portable fueled open-flame devices. Portable open-flame devices fueled by flammable or combustible gases or liquids shall be enclosed or installed in such a manner as to prevent the flame from contacting combustible material.

Exceptions:

1. LP-gas-fueled devices used for sweating pipe joints or removing paint in accordance with Chapter 61.
2. Cutting and welding operations in accordance with Chapter 35.
3. Torches or flame-producing devices in accordance with Section 308.4.
4. Candles and open-flame decorative devices in accordance with Section 308.3.

C This section prohibits the use of portable devices in situations where they might be placed on unstable platforms or where they could be knocked over by human contact. The exceptions list the types of open-flame or heat-producing operations not regulated by this section but regulated elsewhere in the code.

308.1.7 Sky lanterns. A person shall not release or cause to be released an untethered sky lantern.

C Sky lanterns, variously known as Kongming lanterns, Chinese lanterns, fire balloons or fire parachutes, are typically made of paper and contain an open flame used to heat the air inside the device to make it airborne, similar to but on a much smaller scale than a hot air balloon. Once airborne, if these devices are untethered and their movement is thus unrestricted, they are subject to winds and other atmospheric conditions such that the location of the final landfall is completely unknown and uncontrolled by the user. Obviously, uncontrolled open flame devices descending out of the sky have a significant potential to start wildfires and structure fires.

GENERAL REQUIREMENTS

308.1.8 Religious ceremonies. Where, in the opinion of the *fire code official*, adequate safeguards have been taken, participants in religious ceremonies are allowed to carry hand-held candles. Hand-held candles shall not be passed from one person to another while lighted.

C This section has a very narrow application. As stated, only religious ceremonies are covered and the judgment of the fire code official is required for final approval. Prohibiting the passing of lighted candles from person to person is intended to minimize the opportunities for the candles to be dropped where they could become an ignition source for flammable or combustible materials or come in contact with clothing or hair. Spiritual significance may be attached to the use of candles in places of worship; therefore, the local fire code official should work closely with religious groups when enforcing this section.

308.1.9 Aisles and exits. Candles shall be prohibited in areas where occupants stand, or in an *aisle* or *exit*.

C This prohibition is intended to prevent accidents caused by lighted candles being knocked from their holders onto combustible furniture, carpeting or decorative materials. Candles are commonly found at seasonal religious observances where attendance often exceeds the norm. In case of an emergency, people must be able to move through the aisles toward the exits without risking the ignition of clothing, hair or decorations.

308.1.10 Flaming food and beverage preparation. The preparation of flaming foods or beverages in places of assembly and drinking or dining establishments shall be in accordance with Sections 308.1.10.1 through 308.1.10.5.

C The regulations in this section of the code are necessary to give the fire code official guidance in allowing the preparation of flaming food and beverages to be conducted in restaurants in a safe manner. These types of foods, sometimes referred to as “flambé foods,” are prepared by adding a small amount of flammable or combustible liquid in the form of alcoholic beverages to the food and igniting it. Many restaurants prepare popular selected dishes, such as cherries jubilee, brandied peaches and flaming bananas, in this manner, usually tableside within close proximity of the customers. The proximity plus the intentional ignition of flammable liquids in a Group A occupancy make regulation of this process appropriate. Note that the regulations in this section are not intended to address a fuel source, such as Sterno, used to keep food warm.

308.1.10.1 Dispensing. *Flammable or combustible liquids* used in the preparation of flaming foods or beverages shall be dispensed from one of the following:

1. A 1-ounce (29.6 ml) container.
2. A container not exceeding 1-quart (946.5 ml) capacity with a controlled pouring device that will limit the flow to a 1-ounce (29.6 ml) serving.

C These dispensing provisions limit the amount of flammable or combustible liquid being transported around the restaurant for use in flaming food or beverage preparation in order to minimize the fuel potential in a fire incident involving such operations. Item 1 could be referring to a typical shot glass. Item 2 could be referring to a “jigger pourer,” which is a device that fits into the neck of a bottle and is intended to limit the per-pour quantity to 1 ounce.

308.1.10.2 Containers not in use. Containers shall be secured to prevent spillage when not in use.

C Securing the containers used in flaming food or beverage preparation while not in use reduces the likelihood of an accidental spill.

308.1.10.3 Serving of flaming food. The serving of flaming foods or beverages shall be done in a safe manner and shall not create high flames. The pouring, ladling or spooning of liquids is restricted to a maximum height of 8 inches (203 mm) above the receiving receptacle.

C Limiting the height from which flammable or combustible liquids are poured reduces the likelihood of a spill or overpour that might miss the target dish and be ignited by a table candle, smoking materials or another ignition source.

308.1.10.4 Location. Flaming foods or beverages shall be prepared only in the immediate vicinity of the table being serviced. They shall not be transported or carried while burning.

C This section prohibits movement or transport of “flambé foods” while they are burning. This provision reduces the potential for incidents wherein the tray or dish might be dropped or tipped, causing a spill of burning liquid that could lead to a panic reaction by restaurant patrons or, if spilled on a patron, cause serious burn injuries.

308.1.10.5 Fire protection. The person preparing the flaming foods or beverages shall have a wet cloth towel immediately available for use in smothering the flames in the event of an emergency.

C Flaming-food preparation is a cooking hazard much the same as stovetop preparation. An efficient method for extinguishing a stovetop fire in a frying pan or similar utensil is to put the cover on the utensil to exclude oxygen. In flambé preparation, the food is prepared in a relatively small pan or in the dish in which it will be served, in either of which the flames can be easily smothered by a wet towel. Note that this precautionary measure is in addition to the portable fire extinguishers to be provided in accordance with Section 906.

308.2 Permits required. Permits shall be obtained from the *fire code official* in accordance with Section 105.5 prior to engaging in the following activities involving open flame, fire and burning:

1. Use of a torch or flame-producing device to remove paint from a structure.
2. Use of open flame, fire or burning in connection with Group A or E occupancies.
3. Use or operation of torches and other devices, machines or processes liable to start or cause fire in or on wildfire risk areas.

C This section establishes the authority of the fire code official to control the hazards of using open flames through the permitting process. The restrictions here do not prohibit the use of open-flame devices in the listed activities, but they do allow the fire code official to inspect plans and ongoing activities to make certain they are safe for the occupancy in which they are held. Permit applicants should be required to demonstrate that their regulated activities comply with the intent of the code before the permit is issued. See the commentary to Section 105.5 for a general discussion of operations requiring an operational permit and Sections 105.5.37 and 105.5.38 for a discussion of specific operational permits for the indicated regulated activities. The permit process also serves to notify the fire department of the increased hazard at particular locations.

308.3 Group A occupancies. Open-flame devices shall not be used in a Group A occupancy.

Exceptions:

1. Open-flame devices are allowed to be used in the following situations, provided that *approved* precautions are taken to prevent ignition of a combustible material or injury to occupants:
 - 1.1. Where necessary for ceremonial or religious purposes in accordance with Section 308.1.8.
 - 1.2. On stages and platforms as a necessary part of a performance in accordance with Section 308.3.2.
 - 1.3. Where candles on tables are securely supported on substantial noncombustible bases and the candle flames are protected.
2. Heat-producing equipment complying with Chapter 6 and the *International Mechanical Code*.
3. Gas lights are allowed to be used provided that adequate precautions satisfactory to the *fire code official* are taken to prevent ignition of combustible materials.

C The use of open-flame devices in Group A occupancies where large numbers of people gather for entertainment, instruction, food or drink consumption, deliberation, awaiting transportation or social or religious functions increases the likelihood of the occupants coming into contact with these devices and is, therefore, prohibited. Safe alternatives to open-flame devices should be used where practical, especially in restaurants and other assembly occupancies where the focus is on atmosphere rather than symbolism or religious significance.

Exception 1 refers to Sections 308.1.8 and 308.3.2 for use in religious and theatrical settings as well as permitting use on tabletops where properly secured and protected.

Exception 2 refers to building service heat-producing equipment that meets other code requirements.

Exception 3 covers gas lights installed with proper flame safeguards and with the approval of the fire code official. This kind of lighting is often a permanent installation that would be covered by additional code requirements.

308.3.1 Open-flame decorative devices. Open-flame decorative devices shall comply with all of the following restrictions:

1. Class I and Class II liquids and LP-gas shall not be used.
2. Liquid- or solid-fueled lighting devices containing more than 8 ounces (237 ml) of fuel must self-extinguish and not leak fuel at a rate of more than 0.25 teaspoon per minute (1.26 ml per minute) if tipped over.
3. The device or holder shall be constructed to prevent the spillage of liquid fuel or wax at the rate of more than 0.25 teaspoon per minute (1.26 ml per minute) when the device or holder is not in an upright position.
4. The device or holder shall be designed so that it will return to the upright position after being tilted to an angle of 45 degrees (0.79 rad) from vertical.

Exception: Devices that self-extinguish if tipped over and do not spill fuel or wax at the rate of more than 0.25 teaspoon per minute (1.26 ml per minute) if tipped over.
5. The flame shall be enclosed except where openings on the side are not more than 0.375-inch (9.5 mm) diameter or where openings are on the top and the distance to the top is such that a piece of tissue paper placed on the top will not ignite in 10 seconds.
6. Chimneys shall be made of noncombustible materials and securely attached to the open-flame device.

Exception: A chimney is not required to be attached to any open-flame device that will self-extinguish if the device is tipped over.
7. Fuel canisters shall be safely sealed for storage.
8. Storage and handling of *combustible liquids* shall be in accordance with Chapter 57.
9. Shades, where used, shall be made of noncombustible materials and securely attached to the open-flame device holder or chimney.

GENERAL REQUIREMENTS

10. Candelabras with flame-lighted candles shall be securely fastened in place to prevent overturning, and shall be located away from occupants using the area and away from possible contact with drapes, curtains or other combustibles.

C This class of open-flame devices includes items such as wall-mounted candles or torch sconces, bug-repellant candles in glass jars or metal cans, tabletop candles and oil lamps, freestanding torch holders and candelabras. The criteria for the use of this kind of device are all intended to enhance safety.

308.3.2 Theatrical performances. Where *approved*, open-flame devices used in conjunction with theatrical performances are allowed to be used where adequate safety precautions have been taken in accordance with NFPA 160.

C Theatrical performances typically occur on stages and involve large quantities of combustible materials. Hazards associated with stages can include combustible scenery and lighting suspended overhead; scenic elements, contents and acoustical treatment on the back and sides of the stage; workshops, scene docks and dressing rooms located around the stage perimeter, and storage areas and property rooms located underneath the stage. Because of the inherent dangers associated with the introduction of open flames into such a fuel-rich environment, the use of open-flame devices in theatrical performances requires review, evaluation and the approval of the fire code official on a case-by-case basis and must be safeguarded in accordance with the provisions of NFPA 160. For further discussion of the special hazard nature of stages, see the commentary to IBC Section 410.

308.4 Group R occupancies. Open flame, fire and burning in Group R occupancies shall comply with the requirements of Sections 308.1 through 308.1.7 and Section 308.4.1.

C This section establishes the applicability of the indicated requirements of Section 308 to open flame activities in Group R occupancies.

308.4.1 Group R-2 dormitories. Candles, incense and similar open-flame-producing items shall not be allowed in *sleeping units* in Group R-2 dormitory occupancies.

C This section provides the fire code official with the needed authority to control the introduction of potential ignition sources into Group R-2 dormitory occupancies and help to stop dorm fires before lives are endangered and potentially lost. Fire represents a significant risk to life and property in dormitory occupancies, particularly at residential schools, colleges and universities. The large number of young people living in close proximity to one another creates the potential for a relatively small fire to have serious and possibly fatal consequences.

Candle and incense use presents a significant life safety risk in the close confines of a typical residence hall. According to NFPA fire loss data, there is an average of 4,100 fires each year in dormitories, causing \$14 million in direct property damage. The data shows that only 7 percent of fires started in the bedroom, but these fires were responsible for 16 percent of injuries and 24 percent of direct property damage, which makes them the most costly type of fire in a residence hall. Prohibiting the use of candles, incense and similar open flame items reduces the potential ignition sources for a fire, providing for a safer living environment.

SECTION 309—POWERED INDUSTRIAL TRUCKS AND EQUIPMENT

309.1 General. Powered industrial trucks and similar equipment including, but not limited to, floor scrubbers and floor buffers, shall be operated and maintained in accordance with Sections 309.2 through 309.7.

C This section establishes the fire safety requirements for control of the hazards associated with powered industrial trucks and industrial equipment. Because these trucks or equipment may have either battery-powered electric motors or internal combustion engines using liquid or gaseous fuel, Sections 309.2 through 309.7 cover the fire safety aspects of both.

309.2 Use in hazardous (classified) locations. Powered industrial trucks used in areas designated as hazardous (classified) locations in accordance with NFPA 70 shall be *listed* and *labeled* for use in the environment intended in accordance with NFPA 505.

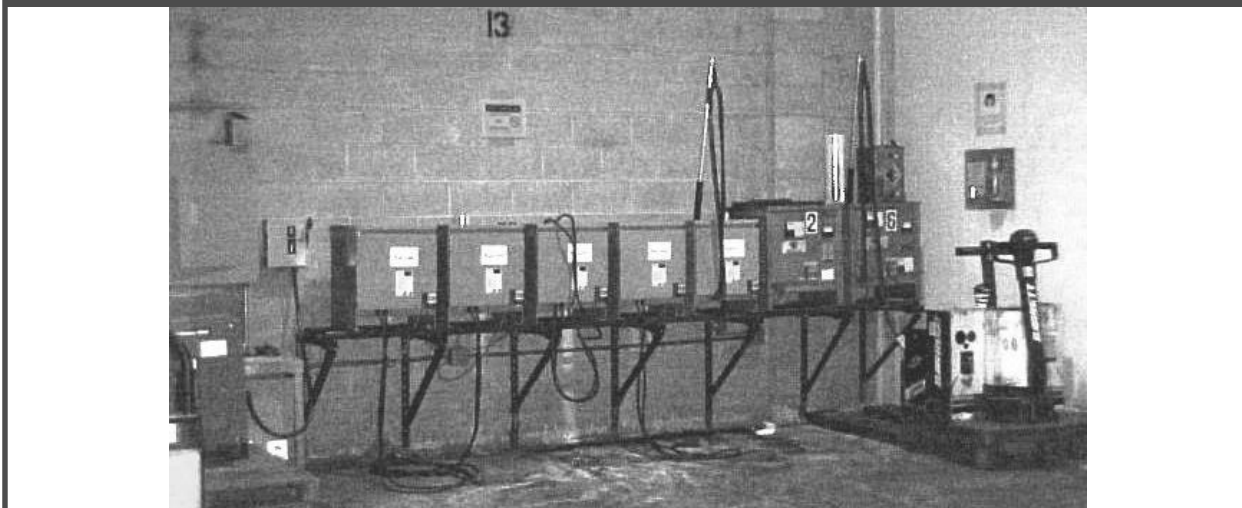
C Powered industrial trucks used in hazardous locations as defined by NFPA 70 must meet the requirements of NFPA 505.

309.3 Battery chargers. Battery chargers shall be of an *approved* type. Combustible storage shall be kept not less than 3 feet (915 mm) from battery chargers. Battery charging shall not be conducted in areas open to the public.

C Battery chargers offer several safety challenges if they are not properly housed and operated. A battery that is connected to the charger with the poles reversed can explode when charging power is turned on, which could result in corrosive liquid being sprayed over the charger room and anyone who happens to be in it. Aside from the possibility of serious bodily injury from flying debris or corrosive spray, eye damage from the spray is a critical consideration. There is also the possibility of energized charging leads shorting, arcing or fusing and causing sparks to become an ignition source for combustibles if they are not kept out of the area.

When a charger is properly connected and energized, the charging process causes generation of hydrogen and oxygen gases as well as acid or alkali fumes. These gases must be vented to prevent them from reaching ignitable or detonable levels (see commentary, Section 309.4). A typical battery charger arrangement is illustrated in Commentary Figure 309.3. Note that these battery chargers are outside of the scope of Section 608 and thus not subject to its provisions.

Commentary Figure 309.3—Battery Chargers for Powered Industrial Trucks



309.4 Ventilation. Ventilation shall be provided in an *approved* manner in battery-charging areas to prevent a dangerous accumulation of *flammable gases*.

C Charging lead-acid or nickel iron batteries is a process of electrolysis in which oxides created by operation of the battery are reduced to metal and redeposited on the electrode plates. The process results in the rejuvenation of the electrolyte in the battery and the emission of both oxygen and hydrogen gases as well as corrosive fumes. If these gases are allowed to accumulate in an enclosed space, they could eventually reach an ignitable or detonable level. Section 1207.6.1 and the applicable provisions of the *International Mechanical Code*® (IMC®) describe one method of ventilation used to carry off and dilute the concentrations of hazardous gases. See the commentary to Section 1207.6.1 and its subsections.

309.5 Fire extinguishers. Battery-charging areas shall be provided with a fire extinguisher complying with Section 906 having a minimum 4-A:20-B:C rating within 20 feet (6096 mm) of the battery charger.

C Because of the electrical hazards associated with the battery-charging operation, the fuel load presented by the plastic battery cases and other area contents and the potential for the presence of gases in the room, an appropriately sized portable fire extinguisher must be located within the battery-charging area. The extinguisher must be accessible with minimum travel.

309.6 Refueling. Powered industrial trucks using liquid fuel, LP-gas or hydrogen shall be refueled outside of buildings or in areas specifically *approved* for that purpose. Fixed fuel-dispensing equipment and associated fueling operations shall be in accordance with Chapter 23. Other fuel-dispensing equipment and operations, including cylinder exchange for LP-gas-fueled vehicles, shall be in accordance with Chapter 57 for *flammable* and *combustible liquids* or Chapter 61 for LP-gas.

C Because of the hazards associated with liquid fuel spills and gaseous fuel discharges, this section requires that powered industrial trucks be refueled outside where the vapors or gas can be readily dissipated. This section also allows the alternative of fueling inside of buildings where safeguards mitigate the hazards of the operation and the location is specifically approved for the use by the fire code official. Where fueling with liquid or gaseous fuel is done from fixed equipment, compliance with Chapter 23 is required for both the fueling equipment and its operation because of the proven safeguards for such operations that are included in that chapter. Similarly, fueling equipment and operations for flammable or combustible liquids or LP-gas utilizing nonfixed equipment must be conducted in accordance with the applicable provisions of the appropriate material-specific chapter.

309.7 Repairs. Repairs to fuel systems, electrical systems and repairs utilizing open flame or welding shall be done in *approved* locations outside of buildings or in areas specifically *approved* for that purpose.

C Repairs that could create ignition sources or a fuel load for an ignition source must be done in indoor or outdoor locations that are designed specifically for vehicle repairs and that have been approved by the fire code official.

SECTION 310—SMOKING

310.1 General. The smoking or carrying of a lighted pipe, cigar, cigarette or any other type of smoking paraphernalia or material is prohibited in the areas indicated in Sections 310.2 through 310.8.

C This section states that smoking is prohibited in those areas designated in Sections 310.2 and 310.8. Before entering an area posted with “No Smoking” signs, anyone carrying a lighted smoking product or device must extinguish the smoking material and properly dispose of the ashes and other residue. Suitable ashtrays should be available at the entry to “No Smoking” areas for disposal of the prohibited smoking materials.

GENERAL REQUIREMENTS

310.2 Prohibited areas. Smoking shall be prohibited where conditions are such as to make smoking a hazard, and in spaces where flammable or combustible materials are stored or handled.

Exception: In Group I-2 occupancies, patients shall be permitted to smoke in designated patient care areas based on the clinical needs of the patient.

C Smoking can be prohibited wherever it would be a hazard in the judgment of the fire code official. The exception allows smoking in certain care areas by patients or residents in Group I-2 occupancies. Psychiatric allowances are also included in clinical needs to manage smoking, particularly among behavioral health addiction patients. Use of the term “designated” enables the facility to establish specific, limited areas for this function and enables the care provider to put the patient in the correct environment.

310.2.1 Group I-2. In Group I-2 occupancies, smoking shall be prohibited in patient care areas or where oxygen is used, stored or handled.

C This section emphasizes that, regardless of the exception to Section 310.2, smoking is not allowed for any reason in any areas where oxygen is used, stored or handled because of the increased hazard of ignition and accelerated burning.

310.3 “No Smoking” signs. The *fire code official* is authorized to order the posting of “No Smoking” signs or the international symbol for no smoking in a conspicuous location in each structure or location in which smoking is prohibited. The content, lettering, size, color and location of required “No Smoking” signs shall be *approved*.

Exception: “No Smoking” signs are not required in interior locations of the facility where signs are displayed at all major entrances into the facility.

C The fire code official is not only authorized to designate where “No Smoking” signs are to be posted but is also responsible for establishing the specification for all aspects of those signs. A typical sign design is shown in Commentary Figure 310.3.

The exception acknowledges that it is redundant and unnecessary to also require “No Smoking” signs to be posted throughout a facility that does not permit smoking anywhere and has such signs displayed at all major entrances.

Commentary Figure 310.3—No Smoking Sign



310.4 Removal of signs prohibited. A posted “No Smoking” sign shall not be obscured, removed, defaced, mutilated or destroyed.

C Posted signs must remain in the locations designated by the fire code official and be readable at all times. Many, if not all, jurisdictions establish penalties for removing, obscuring, defacing or mutilating official signs.

310.5 Compliance with “No Smoking” signs. Smoking shall not be permitted nor shall a person smoke, throw or deposit any lighted or smoldering substance in any place where “No Smoking” signs are posted.

C Penalties are usually imposed for violating “no smoking” prohibitions. Smoking must be confined to approved areas, and discarded smoking materials must be deposited only in approved ashtrays or receptacles. Violation of this provision customarily constitutes a misdemeanor. Prosecution of misdemeanor offenses should be coordinated with the jurisdiction’s legal counsel. Offense and penalty clauses, such as those included in Section 113, are required in order to prosecute infractions (see commentary, Section 113.3).

310.6 Ash trays. Where smoking is permitted, suitable noncombustible ash trays or match receivers shall be provided on each table and at other appropriate locations. In Group I-2 occupancies, noncombustible metal containers with self-closing covers shall be provided in areas where smoking is permitted.

C Where smoking is permitted, smokers must have available a noncombustible ashtray or other receptacle where smoking materials can be safely deposited. These receptacles must be located on each table and other locations throughout the smoking area so that the smoker does not have to look for a place to discard used materials. Group I-2 occupancies that allow smoking in specific areas in accordance with the exception to Section 310.2 are to also provide noncombustible metal containers that have self-closing covers in each area.

310.7 Burning objects. Lighted matches, cigarettes, cigars or other burning object shall not be discarded in such a manner that could cause ignition of other combustible material.

C Lighted matches, burning tobacco products and all other burning objects must be deposited in ashtrays or other approved noncombustible containers to separate these ignition sources from any potential fuel loads.

310.8 Hazardous environmental conditions. Where the *fire code official* determines that hazardous environmental conditions necessitate controlled use of smoking materials, the ignition or use of such materials in mountainous, brush-covered or forest-covered areas or other designated areas is prohibited except in *approved* designated smoking areas.

C This section gives the fire code official the authority to establish “no smoking” areas whenever and wherever environmental conditions are considered hazardous. This can include prohibiting burning dry grasses and leaves where lack of rain has rendered the environment dangerously dry or burning of specific materials where the smoke plume from the fire would endanger the health or welfare of a population. The possibilities for determination of hazardous conditions are too numerous to itemize. The judgment of the fire code official, with the advice of other officials being sought as needed, is the determining factor. The fire code official can also exercise the discretion of designating areas in which smoking is permitted.

SECTION 311—VACANT PREMISES

311.1 General. Temporarily unoccupied buildings, structures, premises or portions thereof, including tenant spaces, shall be safeguarded and maintained in accordance with Sections 311.1.1 through 311.6.

C Vacant buildings or portions of buildings that are open at doors or windows pose fire safety and criminal trespass hazards to a community and are correctly declared to be unsafe buildings in Section 115.1.1. Such premises are often called an “attractive nuisance” to neighborhood children who may enter them to play or to other persons who may enter seeking shelter from the elements or to engage in potential criminal activities.

311.1.1 Abandoned premises. Buildings, structures and premises for which an *owner* cannot be identified or located by dispatch of a certificate of mailing to the last known or registered address, which persistently or repeatedly become unprotected or unsecured, which have been occupied by unauthorized persons or for illegal purposes, or which present a danger of structural collapse or fire spread to adjacent properties shall be considered to be abandoned, declared unsafe and abated by demolition or rehabilitation in accordance with the *International Property Maintenance Code* and the *International Building Code*.

C This section establishes the authority to demolish or rehabilitate properties that pose a variety of public safety hazards when the owners of the property cannot be located by customary legal means. Because demolition and rehabilitation are regulated by the IBC, *International Property Maintenance Code*® (IPMC®) and applicable state laws, any action taken must be a carefully coordinated effort by all affected code officials in close relationship with the jurisdiction’s legal counsel.

311.1.2 Tenant spaces. Storage and lease plans required by this code shall be revised and updated to reflect temporary or partial vacancies.

C This section is intended to keep storage and lease plans up to date so that the fire service will always have a complete picture of the kinds of hazards it might face in case of a fire, including vacant, unattended spaces.

311.2 Safeguarding vacant premises. Temporarily unoccupied buildings, structures, premises or portions thereof shall be secured and protected in accordance with Sections 311.2.1 through 311.2.3.

C This section lists a number of problems that commonly occur where buildings or portions of buildings remain vacant for long periods of time. It covers concerns for security, fire protection and fire separation in vacant spaces.

311.2.1 Security. Exterior and interior openings open to other tenants or unauthorized persons shall be boarded, locked, blocked or otherwise protected to prevent entry by unauthorized individuals. The *fire code official* is authorized to placard, post signs, erect barrier tape or take similar measures as necessary to secure public safety.

C Unauthorized or illegal activities in vacant buildings can lead to the presence of unanticipated fire loads susceptible to ready ignition and rapid fire spread, thus increasing the hazard to adjoining properties or spaces and fire department personnel (see commentary, Section 115.1.1). This section requires securing the openings of vacant buildings or spaces against unauthorized entry by any of the methods listed or by other equally effective means approved by the fire code official.

GENERAL REQUIREMENTS

cial and provides that appropriate signage or other readily visible warnings to the public may be posted. For a suggested standard for boarding windows and doors, see IPMC Appendix A.

311.2.2 Fire protection. Fire protection systems shall be maintained in an operable condition at all times.

Exceptions:

1. Where the premises have been cleared of all combustible materials and debris and, in the opinion of the *fire code official*, the type of construction, *fire separation distance* and security of the premises do not create a fire hazard.
2. Where *approved* by the *fire code official*, buildings that will not be heated and where *fire protection systems* will be exposed to freezing temperatures, fire alarm and *automatic sprinkler systems* are permitted to be placed out of service and standpipes are permitted to be maintained as dry systems (without an automatic water supply), provided that the building does not have contents or storage, and windows, doors and other openings are secured to prohibit entry by unauthorized persons.
3. Where *approved* by the *fire code official*, fire alarm and *automatic sprinkler systems* are permitted to be placed out of service in seasonally occupied buildings: that will not be heated; where fire protection systems will be exposed to freezing temperatures; where *fire areas* do not exceed 12,000 square feet (1115 m²); and that do not store motor vehicles or hazardous materials.

C The basic requirement of this section is clearly stated. The on-site fire protection systems must be maintained whether the property is occupied or vacant. The systems would be subject to the same inspections in either case.

Exception 1 gives the fire code official the authority to determine whether vacant premises pose a significant hazard and lists criteria they can use in making that determination.

Exception 2 recognizes that systems located in unheated premises in cold climates could be rendered inoperable by freezing, and authorizes permission to disable those systems where the security of the premises and fire separation arrangements meet the requirements of Sections 311.2.1 and 311.2.3. Since an inoperative fire protection system has a direct effect on the tactical approaches and operations of the fire department, this exception is applicable only where the fire chief of the jurisdiction has approved the securing of the system.

Exception 3 permits a seasonally occupied building that would be subject to freezing temperatures to be treated like a storage occupancy rather than an occupancy having fire protection systems; mainly assembly occupancies. It incorporates the limitation of 12,000 square feet as the maximum fire area since this is the maximum size a Group S-1 fire area can be without sprinkler protection. Even though a Type VB construction building can only be 9,000 square feet as limited by IBC Table 506.2, allowing up to 12,000 square feet in line with the sprinkler requirements is considered to be more appropriate since these buildings generally have the access around them and the heating system is not operational. Further, the limitation of vehicles and hazardous materials is done to address areas where sprinkler protection would be required at a lower threshold. Even though the thresholds have been focused on sprinkler systems, the disabling of fire alarm systems and smoke alarms is less of an issue in seasonally occupied buildings since sprinkler thresholds are mainly based on the type and ability of occupants.

311.2.3 Fire separation. *Fire-resistance-rated* partitions, *fire barriers* and *fire walls* separating vacant tenant spaces from the remainder of the building shall be maintained. Openings, joints and penetrations in *fire-resistance-rated* assemblies shall be protected in accordance with Chapter 7.

C Fire-resistance-rated construction separating vacant spaces from the remainder of the building must be maintained to the satisfaction of the fire code official. The requirements for openings, joints and penetrations are covered in Chapter 7.

311.3 Removal of combustibles. Persons owning, or in charge or control of, a vacant building or portion thereof, shall remove therefrom all accumulations of combustible materials, flammable or combustible waste or rubbish and shall securely lock or otherwise secure doors, windows and other openings to prevent entry by unauthorized persons. The premises shall be maintained clear of waste or hazardous materials.

Exceptions:

1. Buildings or portions of buildings undergoing additions, *alterations*, repairs or change of occupancy in accordance with the *International Building Code*, where waste is controlled and removed as required by Section 304.
2. Seasonally occupied buildings.

C Property owners, their agents and persons leasing vacant spaces are responsible for preventing accumulations of flammable or combustible materials as well as for securing the vacant space against entry by unauthorized persons.

Exception 1 covers building situations in which larger amounts of flammable or combustible waste would reasonably be generated and cites code references to cover requirements in those situations.

Exception 2 allows reasonable accumulations of flammable or combustible materials in spaces that are occupied seasonally. For example, this exception would allow unattended off-season storage of stock for sale.

311.4 Removal of hazardous materials. Persons owning or having charge or control of a vacant building containing hazardous materials regulated by Chapter 50 shall comply with the facility closure requirements of Section 5001.6.

C This section gives the fire code official the authority to require property owners, their agents and their tenants to submit a facility closure plan, and make sure combustible and hazardous materials are removed from the premises. The requirements for the facility closure plan are given in Section 5001.6 (see Section 407.7).

311.5 Placards. Any vacant or abandoned buildings or structures determined to be unsafe pursuant to Section 115 relating to structural or interior hazards shall be marked as required by Sections 311.5.1 through 311.5.5.

C Vacant or abandoned buildings are often of questionable structural integrity and have a very high probability of intentionally set fires. When fires occur in these buildings, they present a host of unusual hazards to firefighters. Since the buildings are uninhabited, fires may develop for significant periods of time before they are detected and reported.

In December 1999, six professional firefighters died after they became lost in a six-floor, maze-like, cold-storage and warehouse building while searching for two reportedly trapped people and fire extension. In the wake of this tragedy, the National Institute for Occupational Safety and Health (NIOSH) conducted a Fatality Assessment and Control Evaluation (FACE) as part of its Fire Fighter Fatality Investigation and Prevention Program and issued its findings in report FACE-99-F47 in September 2000. NIOSH investigators concluded that, to reduce the risk of similar occurrences, fire departments should, among other recommendations, implement a program to “identify dangerous vacant buildings by affixing warning placards to entrance doorways or other openings where firefighters may enter.”

In order to improve firefighter safety when responding to incidents in buildings that have structural safety issues and have been declared to be unsafe due to being vacant or abandoned, this section establishes an information placarding system that will assist incident commanders in making personnel and equipment deployment decisions.

311.5.1 Placard location. Placards shall be applied on the front of the structure and be visible from the street. Additional placards shall be applied to the side of each entrance to the structure and on penthouses.

C Prominent positioning of warning placards where they can be readily seen by first-in fire companies or other emergency responders will assist in the initial incident size-up and resource assignments.

311.5.2 Placard size and color. Placards shall be 24 inches by 24 inches (610 mm by 610 mm) minimum in size with a red background, white reflective stripes and a white reflective border. The stripes and border shall have a 2-inch (51 mm) minimum stroke.

C In order to be truly effective in conveying the critical information about a building’s structural stability, placards must be fabricated so as to be in bold contrast to their mounting surface. The required reflectivity will enhance placard visibility under low ambient light conditions where flashlights, spotlights or emergency vehicle warning lights may be the only light-emitting source.

311.5.3 Placard date. Placards shall bear the date of their application to the building and the date of the most recent inspection.

C Including the date of placard posting and the most recent inspection information on the placard will assist the incident commander in verifying the validity and currency of the building’s status.

311.5.4 Placard symbols. The design of the placards shall use the following symbols:

1. ☐ This symbol shall mean that the structure had normal structural conditions at the time of marking.
2. ☐ This symbol shall mean that structural or interior hazards exist and interior firefighting or rescue operations should be conducted with extreme caution.
3. ☒ This symbol shall mean that structural or interior hazards exist to a degree that consideration should be given to limit firefighting to exterior operations only, with entry only occurring for known life hazards.
4. Vacant marker hazard identification symbols: The following symbols shall be used to designate known hazards on the vacant building marker. They shall be placed directly above the symbol.
 - 4.1. R/O—Roof open.
 - 4.2. S/M—Stairs, steps and landing missing.
 - 4.3. F/E—Avoid fire escapes.
 - 4.4. H/F—Holes in floor.

C The placard symbols required by this section are intended to provide clear information regarding the level of structural hazard facing personnel in the conduct of rescue and firefighting operations as of the inspection date on the placard. Upon considering the degree of hazards, incident commanders can restrict operations to strictly defensive or cautiously offensive.

311.5.5 Informational use. The use of these symbols shall be informational only and shall not in any way limit the discretion of the on-scene incident commander.

C This section simply states that the placards are only an operational tool and the scene incident commander is the final authority in applying the information provided by the placards to the situation at hand.

GENERAL REQUIREMENTS

311.6 Unoccupied tenant spaces in mall buildings. Unoccupied tenant spaces in covered and open mall buildings shall be:

1. Kept free from the storage of any materials.
2. Separated from the remainder of the building by partitions of not less than 0.5-inch-thick (12.7 mm) gypsum board or an *approved* equivalent to the underside of the ceiling of the adjoining tenant spaces.
3. Without doors or other access openings other than one door that shall be kept key locked in the closed position except during that time when opened for inspection.
4. Kept free from combustible waste and be broomswept clean.

C This section is primarily concerned with the hazards posed by tenant spaces that are not in use or that are not under the supervision of employees. These spaces are more likely to be targets of vandalism and incendiary activity. Generally, a fire can grow unnoticed in such spaces as a result of the lack of supervision or activity in the space. To reduce these risks, this section contains several requirements that focus on reducing the fire ignition, growth and spread potential by limiting combustibles in the space, securing the space through the use of locks and installing fire separations constructed of 1/2-inch-thick (12.7 mm) gypsum or similar materials. Note that mall buildings are required to be sprinklered throughout and that the systems in the tenant spaces must be independent of the mall area.

SECTION 312—VEHICLE IMPACT PROTECTION

312.1 General. Vehicle impact protection required by this code shall be provided by posts that comply with Section 312.2 or by other *approved* physical barriers that comply with Section 312.3.

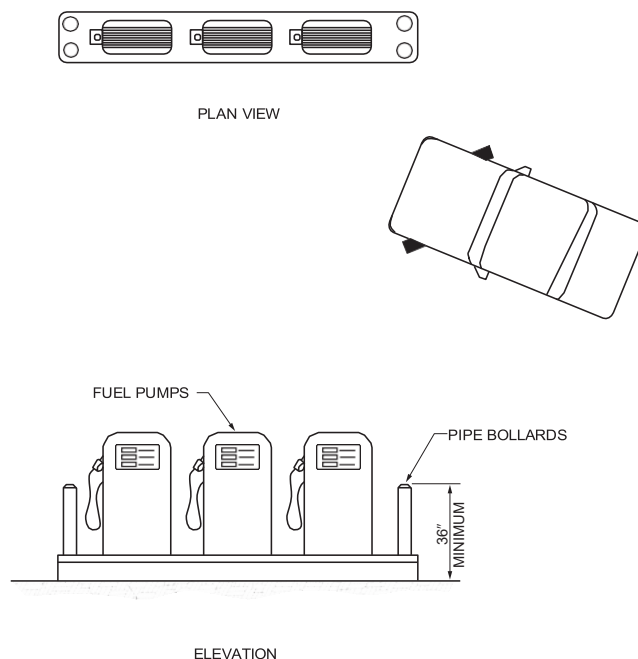
C This section applies to those locations where a moving vehicle could strike a piece of equipment that contains fuel or is fuel fired. These applications include motor fuel-dispensing facilities, above-ground storage tanks and repair garages as well as other locations in which gas- or oil-fired equipment or appliances could be installed where they would be in harm's way. Additional requirements for equipment protection are contained in Chapter 3 of the IMC and Section 305 of the *International Fuel Gas Code*® (IFGC®). Acceptable impact protection is dependent upon location and physical arrangement of the equipment requiring protection and is not limited to guard posts only. As indicated in Section 312.3, protection could consist of walls, barricades, guardrails or elevated locations that provide the requisite level of protection and are approved by the fire code official.

312.2 Posts. Guard posts shall comply with all of the following requirements:

1. Constructed of steel not less than 4 inches (102 mm) in diameter and concrete filled.
2. Spaced not more than 4 feet (1219 mm) between posts on center.
3. Set not less than 3 feet (914 mm) deep in a concrete footing of not less than a 15-inch (381 mm) diameter.
4. Set with the top of the posts not less than 3 feet (914 mm) above ground.
5. Located not less than 3 feet (914 mm) from the protected object.

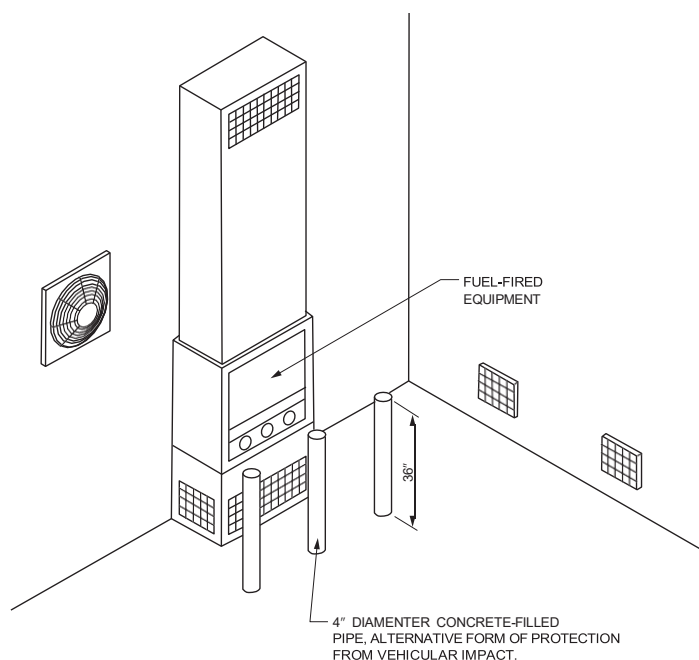
C This section lists five requirements that guard posts must satisfy. Typical installations of posts in service stations and other locations are shown in Commentary Figures 312.2(1) and 312.2(2). These guard posts are designed to resist impact from vehicles moving at low speeds, as they would be when pulling up to a fuel pump at a motor fuel-dispensing facility or into a service area in an indoor service facility or repair garage.

Commentary Figure 312.2(1)—Pump Island Protection



For SI: 1 inch = 25.4 mm.

Commentary Figure 312.2(2)—Guard Post Installation in Public Repair Garage



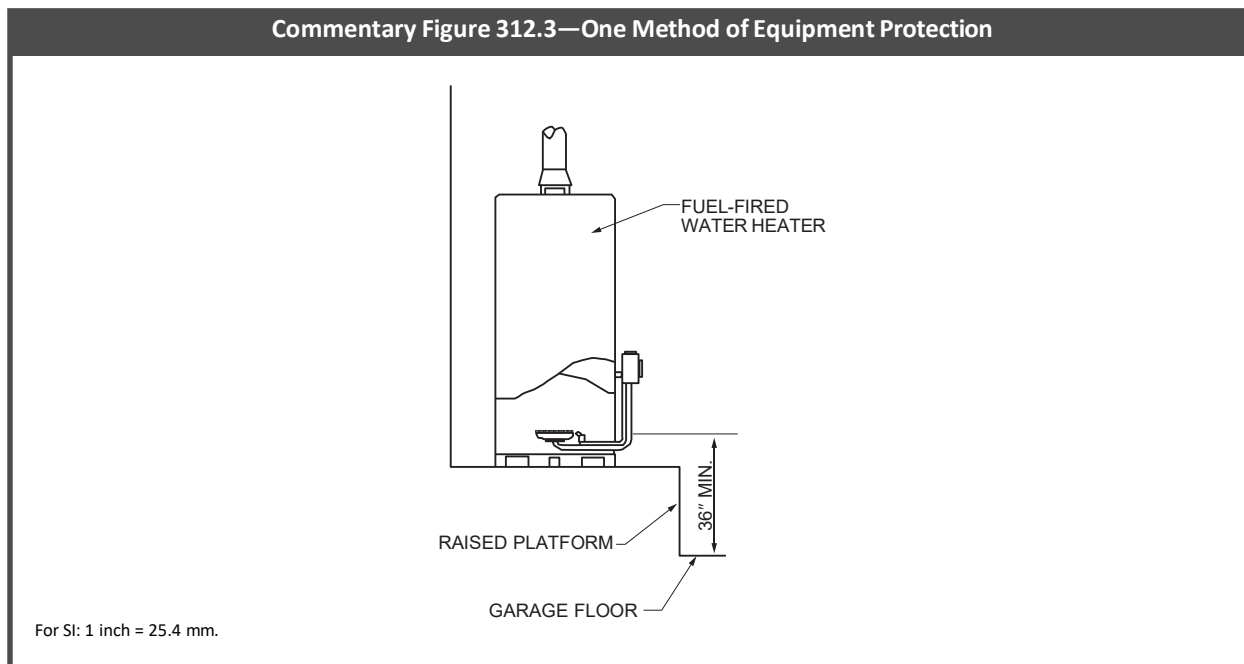
For SI: 1 inch = 25.4 mm.

312.3 Other barriers. Barriers, other than posts specified in Section 312.2, that are designed to resist, deflect or visually deter vehicular impact commensurate with an anticipated impact scenario shall be permitted where *approved*.

C This section provides a true performance option to Section 312.2. The text deliberately establishes a broad set of goals that must be achieved by the designer to fit a site-specific application, and the requirement places the onus on the designer to demonstrate selection of a satisfactory design scenario and a suitable solution to achieve approval of the fire code official. Although it could be argued that Section 312.3 could simply be deleted in favor of relying on Section 104.2.3 (alternate

GENERAL REQUIREMENTS

materials and methods), it made more sense to include the suggested guidance in Section 312.3 of the code. Previously, the 12,000-pound resistance force prescribed by this section was actually a static load (i.e., a load with no associated impact velocity or acceleration) rather than an impact load. Without knowing an intended impact velocity, the kinetic energy resistance for a barrier cannot be accurately calculated. It is more appropriate for a performance requirement to accommodate determination of a suitable vehicle weight and impact speed to provide protection from moving vehicles as a design basis. Barriers other than posts could include walls, barricades or elevated locations for equipment (see Commentary Figure 312.3).



SECTION 313—FUELED EQUIPMENT

313.1 General. Fueled equipment including, but not limited to, motorcycles, mopeds, lawn-care equipment and portable generators shall not be stored, operated or repaired within a building.

Exceptions:

1. Buildings or rooms constructed for such use in accordance with the *International Building Code*.
2. Where allowed by Section 314.
3. Storage of equipment utilized for maintenance purposes is allowed in *approved* locations where the aggregate fuel capacity of the stored equipment does not exceed 10 gallons (38 L) and the building is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

C The restrictions in this section are similar to those in Section 314.4, but this section also regulates the storage, operation or repair of portable liquid- or gas-fueled generators, and certain vehicles (see commentary, Section 314.4). It has been documented that multiple fatalities occurred, related to natural disasters, as the result of the improper use and location of portable generators inside buildings. This section clarifies that portable generators are not allowed to be used inside buildings except when utilized in accordance with one of the exceptions.

Exception 1 recognizes the increased safety afforded when these uses are isolated from other parts of the building with fire-resistance-rated construction in accordance with the IBC. Exception 2 defers to the other section of the code that also regulates fueled equipment covered by this section. Exception 3 recognizes the enhanced level of protection provided by an approved automatic sprinkler system by allowing fueled equipment to be stored inside in a location approved by the fire code official, up to an aggregate quantity of 10 gallons (38 L) of fuel. This exception provides consistency with Section 5704.3.4.4 and, therefore, is applicable to flammable and combustible liquid fuels only; liquefied gas fuels are not included. Note also that the fuel quantities allowed by this exception are not to be counted when determining maximum allowable quantities in accordance with Table 5003.1.1(5).

313.1.1 Removal. The *fire code official* is authorized to require removal of fueled equipment from locations where the presence of such equipment is determined by the *fire code official* to be hazardous.

C This section gives the fire code official the authority to conduct inspections to determine whether vehicle operation, repair and storage in buildings comply with the requirements of this section and, if they do not, the authority to order the removal of fueled equipment from the building as a means of eliminating the hazard.

313.2 Group R occupancies. Vehicles powered by *flammable liquids*, Class II *combustible liquids* or compressed *flammable gases* shall not be stored within the living space of Group R buildings.

C Prohibiting storage of fuel-burning vehicles recognizes the hazards associated with having significant quantities of flammable or combustible liquids or compressed gases in inhabited spaces. Most vehicles that use liquid fuels have tanks that are not pressurized and are not vapor tight. Even a small leak over time can build to dangerous levels in enclosed spaces. For these reasons, this section prohibits the storage of gas- or liquid-fueled vehicles and equipment within the dwelling units or sleeping units of Group R buildings, including their associated balconies.

SECTION 314—INDOOR DISPLAYS

314.1 General. Indoor displays constructed within any occupancy shall comply with Sections 314.2 through 314.4.

C Indoor displays of merchandise and the display of all manner of vehicles inside of buildings can create a number of hazards to building occupants, including blocked egress and rapid fire buildup. This section describes reasonable measures to reduce the hazards associated with indoor displays without prohibiting them.

314.2 Fixtures and displays. Fixtures and displays of goods for sale to the public shall be arranged so as to maintain free, immediate and unobstructed access to *exits* as required by Chapter 10.

C The reason for maintaining free and unobstructed access to exits in public shopping spaces is, of course, personal safety in times of emergency. Chapter 10 contains the requirements, criteria and guidelines for this purpose.

314.3 Highly combustible goods. The display of highly combustible goods, including but not limited to fireworks, *flammable* or *combustible liquids*, liquefied flammable gases, oxidizing materials, pyroxylin plastics and agricultural goods, in main *exit access aisles*, *corridors*, covered and open malls, or within 5 feet (1524 mm) of entrances to *exits* and exterior exit doors is prohibited where a fire involving such goods would rapidly prevent or obstruct egress.

C As stated in Chapter 10, all elements of the means of egress of any occupancy open to the public must be kept clear of obstructions and other hazards that could prevent the occupants from exiting the premises quickly in an emergency. Displaying the hazardous materials itemized in this section where their involvement in a fire would block exit pathways is prohibited for this reason. The hazards associated with each of the materials mentioned in the section are discussed in the commentary to Chapters 50 through 67.

314.4 Vehicles. Liquid-fueled or gaseous-fueled vehicles, aircraft, boats or other motorcraft shall not be located indoors except as follows:

1. The engine starting system is made inoperable or ignition batteries are disconnected except where the *fire code official* requires that the batteries remain connected to maintain safety features.
2. Fuel in fuel tanks does not exceed any of the following:
 - 2.1. Class I, II and III liquid fuel does not exceed one-quarter tank or 5 gallons (19 L), whichever is less.
 - 2.2. LP gas does not exceed one-quarter tank or 6.6 gallons (25 L), whichever is less.
 - 2.3. CNG does not exceed one-quarter tank or 630 cubic feet (17.8 m³), whichever is less.
 - 2.4. Hydrogen does not exceed one-quarter tank or 2,000 cubic feet (57 m³), whichever is less.
3. Fuel tanks and fill openings are closed and sealed to prevent tampering.
4. Vehicles, aircraft, boats or other motorcraft equipment are not fueled or defueled within the building.

C It has become commonplace for covered malls and larger retail stores to have various types of gas- or liquid-fueled vehicles on inside display, such as for promotional events or fire apparatus displays during Fire Prevention Week (see Commentary Figure 314.4). Because the hazards of such displays in a public building are similar to those in residential buildings, Section 314.4 parallels Section 313.1, Exception 2 (see commentary, Section 313.1). New electric and hybrid vehicles have traditional 12V ignition batteries and main batteries for propulsion. This section clarifies in Item 1 that the ignition batteries should be disconnected. The main batteries of these vehicles are typically disconnected when the vehicle ignition batteries are disconnected. Item 2 includes criteria for alternative fuels, requiring that the tanks for each type do not exceed the amounts listed rather than requiring that these vehicles have to completely purge all fuel out of their tanks when they are located at indoor displays. Purging could allow oxygen to enter the tank which, when refueled, may create an explosive atmosphere. In addition, for composite tanks, the liner may be damaged if the internal pressure is reduced to zero. This item specifies the quantities allowed for each alternative fuel with the energy equivalent of 5 gallons of Class 1 liquid fuel (gasoline).

Commentary Figure 314.4—Indoor Car Display



SECTION 315—GENERAL STORAGE

315.1 General. Storage shall be in accordance with Sections 315.2 through 315.6. Outdoor pallet storage shall be in accordance with Sections 315.2 and 315.7.

Exception: Wood and wood composite pallets stored outdoors at pallet manufacturing and recycling facilities and complying with Section 2810.

C This section addresses general safety precautions for the storage of materials, including those regulated elsewhere in the code. Where a material in storage is specifically regulated by another section of the code, that section would apply in addition to any applicable provisions in Section 315. For example, if the stored material were baled wastepaper awaiting recycling, the material-specific provisions of Section 3705 would apply. However, Section 3705 does not specifically regulate the vertical clearance of the stored baled wastepaper to the ceiling or sprinkler heads, nor does it regulate clearance from heaters or heating devices. Accordingly, the provisions of Sections 315.3 and 315.3.1 would apply. Also note that if the storage meets the definition of high-piled combustible storage in Section 202, the applicable provisions of Chapter 32 would apply. Control of combustible waste material is covered in Sections 304.2 and 304.3.

315.2 Permit required. A permit for miscellaneous combustible storage shall be required as set forth in Section 105.5.

C This section establishes the requirement that an operational permit for storage of certain combustible materials must be obtained from the fire code official. Permit fees, if any, must be paid prior to the issuance of the permit. See Section 105 for additional information on permits. Note that a permit is required only for storage of combustible materials. Even though Section 315.3 regulates storage of both combustible and noncombustible materials, a permit is required only for miscellaneous types of combustible materials and then only when the storage volume exceeds 2,500 cubic feet (71 m³), as specified in the permit requirements in Section 105.5.33. The storage of such combustible materials can easily fall into disarray, present a fire load out of proportion to the surroundings and be susceptible to ready ignition and rapid, uncontrolled combustion. Conversely, the orderly storage of mercantile stocks and goods, business records, etc., in piles, on shelving or in casework cabinets, for example, while regulated by Section 315, is not considered to be a hazard requiring a permit.

315.3 Storage in buildings. Storage of materials in buildings shall be orderly and stacks shall be stable. Storage of combustible materials shall be separated from heaters or heating devices by distance or shielding so that ignition cannot occur.

C The first sentence of this section makes it clear that it applies to all storage and that the stacks of stored materials must be stable in order to prevent pile or stack collapse that could obstruct means of egress aisles and impede firefighter access. The second sentence applies only to storage of combustible materials and the need for them to be separated from ignition sources. Throughout the code, the use of fire-resistance-rated construction and spatial separation distances to minimize fire hazards and fire spread is stated as a requirement for a variety of materials. It is not necessary to separate noncombustible materials from ignition sources because there is no combustible material hazard.

315.3.1 Ceiling clearance. Storage shall be maintained 2 feet (610 mm) or more below the ceiling in nonsprinklered areas of buildings or not less than 18 inches (457 mm) below sprinkler deflectors in sprinklered areas of buildings.

Exceptions:

1. The 2-foot (610 mm) ceiling clearance is not required for storage along walls in nonsprinklered areas of buildings.
2. The 18-inch (457 mm) ceiling clearance is not required for storage along walls in areas of buildings equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1, 903.3.1.2 or 903.3.1.3.

C If the space is not equipped with sprinklers, the clearance between the stored materials and the ceiling must be 2 feet (610 mm) to allow manual hose streams to effectively reach the top of a burning pile as well as to project over and beyond adjacent piles to reach burning materials. Where sprinklers are installed, the 18-inch (457 mm) clearance permits timely activa-

tion of the sprinklers and allows unobstructed water distribution over the storage pile. Materials stored too close to sprinklers can not only prevent the heat of a fire from reaching the sprinkler fusible link but also inhibit water from reaching the seat of a fire once the sprinklers are activated.

In sprinklered areas of buildings, the question often arises as to whether storage on shelving can be installed on a wall not directly below sprinklers, with the storage extending above the level of a horizontal plane located 18 inches (457 mm) below the ceiling sprinkler deflector. Typically, storage on wall-mounted shelving functions the same as a wall with respect to its relationship to the automatic sprinklers installed at ceiling level and the relative obstruction it poses. As long as the storage is not directly below the sprinklers, the sprinkler clearance provisions of Section 315.2.1 would not apply. This is consistent with NFPA 13 annex notes on this topic.

Certain newer types of automatic sprinklers, because of their design or operating characteristics, may require greater clearance distances than the 18-inch (457 mm) minimum prescribed in this section. NFPA 13 and the sprinkler manufacturer's data should be consulted for specific information on the characteristics of the many different types of sprinklers that may be installed in a given building.

315.3.2 Means of egress. Combustible materials shall not be stored in *exits* or enclosures for *stairways* and *ramps*. Combustible materials in the *means of egress* during construction, demolition, remodeling or *alterations* shall comply with Section 3310.1.

C As was stated in Section 314.3, all elements of the means of egress must be kept free of obstructions that could block an exit pathway and, thus, jeopardize occupants of the affected space. Chapter 10 offers more guidance on means of egress.

315.3.3 Equipment rooms. Combustible material shall not be stored in boiler rooms, mechanical rooms, elevator machine rooms, electrical equipment rooms or in *fire command centers* as specified in Section 508.1.5.

C This section is intended to keep ignition sources inherent in the use of certain indicated rooms (e.g., fuel-fired equipment, electrical panels) from coming into contact with combustible materials that might be stored in the rooms, whether the rooms are sprinklered or not. See also the commentary to Section 508.1.5 for storage prohibition in fire command centers and Section 605.3 for electrical equipment room clearance requirements and storage prohibition. Note that IMC Chapter 2 defines a "Mechanical equipment/appliance room" as one in which nonfuel-fired mechanical equipment is located. Even though the ignition hazard is reduced in the absence of fuel-fired equipment in such rooms, they must be kept free of combustible storage to allow ready access for firefighters and other authorized personnel to reach easily critical controls, such as power or equipment disconnects, in case of an emergency. Limited supplies of combustible materials related to mechanical room equipment (e.g., air distribution system filters) could be allowed but only in accordance with Section 315 of the code and with the specific approval of the fire code official. For additional discussion of requirements applicable to these and other specific occupancy rooms, see IBC Section 509. Further discussion of the hazards of storage in boiler rooms can be found in IMC Section 304.

315.3.4 Attic, under-floor and concealed spaces. Attic, under-floor and concealed spaces used for storage of combustible materials shall be protected on the storage side as required for 1-hour *fire-resistance-rated* construction. Openings shall be protected by assemblies that are self-closing and are of noncombustible construction or solid wood core not less than 1³/₄ inches (44.5 mm) in thickness. Storage shall not be placed on exposed joists.

Exceptions:

1. Areas protected by *approved automatic sprinkler systems*.
2. Group R-3 and Group U occupancies.

C This section recognizes the reality that attics, crawl spaces and similar unoccupied, concealed spaces in buildings are attractive to building occupants for storage of all kinds of combustible materials. Storage in such unattended and out-of-the-way spaces creates a hazardous condition by introducing a higher fire load to spaces that were neither designed nor intended for such a high-intensity use and in which a fire could rapidly develop unobserved until it had gained a considerable hold on the building. Placing stored combustibles on exposed joists also could hasten collapse of the joists in a fire, which could lead to flaming debris being dropped into the building space below the joists and possible collapse of all or part of the building structure.

The code provides alternatives to using such spaces for storage and, if they are used for storage, how they can be constructed to isolate the higher fire loads created. Since the intent is to protect against a fire in the storage area from endangering the other occupied areas of the building, the required 1-hour fire-resistance rating need only be achieved from the storage side. While any access openings in the 1-hour fire-resistant construction need not be rated, they must be self-closing and of either noncombustible construction or a minimum 1 3/4-inch (44 mm) thickness of solid wood core. Consistent with Section 102.4, any construction in connection with the concealed spaces regulated by this section must be in accordance with the IBC, especially Section 509.

Exception 1 recognizes the efficiency and reliability of automatic sprinklers as a trade-off for 1-hour fire-resistance-rated construction. This exception only requires the sprinkler system in the attic, under-floor or concealed space. Complete sprinkler protection throughout the building is not required in order to be in compliance with the exception.

GENERAL REQUIREMENTS

Exception 2 exempts Group R-3 and U occupancies from the requirements of this section. In Group R-3, the level of familiarity and control exercised by the building occupants is recognized as offsetting the hazards of storage in concealed spaces. Because Group U occupancies are generally unoccupied, the hazards of miscellaneous storage are of little or no consequence to the few occupants that might be in such buildings. For further information on Group U occupancies, see IBC Section 312 and its commentary.

315.4 Outside storage. Outside storage of combustible materials shall not be located within 10 feet (3048 mm) of a *lot line*.

Exceptions:

1. The separation distance is allowed to be reduced to 3 feet (914 mm) for storage not exceeding 6 feet (1829 mm) in height.
2. The separation distance is allowed to be reduced where the *fire code official* determines that hazard to the adjoining property does not exist.

C Outside storage of combustible materials, such as raw materials for production, idle pallets, dunnage and packaging, must be neat and compact. The requirement for a 10-foot (3048 mm) separation is consistent with storage area aisle width requirements throughout the code, often expressed as “one-half the pile height or 10 feet (3048 mm), whichever is greater.” The requirement of this section is consistent with that concept, as is Section 315.3.2, which limits pile height to 20 feet (6096 mm). This section is intended to provide fire suppression access on all sides of storage arrangements and reduce the likelihood of the spread of fire to adjacent properties in the event of a pile collapse. Pile collapses will generally not involve a full-height topple-over of a pile, but rather only a partial collapse. Accordingly, Exception 1 allows a reduction in separation distance where the pile height is substantially less than the separation requirement.

Exception 2 allows the fire code official to grant separation reductions when the combustibles are judged to be no threat to adjoining property. Examples of such conditions could include storage where the combustible materials are enclosed in noncombustible containers, the presence of an impervious property line barrier or the provision of fixed fire protection equipment, such as deluge monitors, especially designed for rapid fire suppression and exposure protection.

315.4.1 Storage beneath overhead projections from buildings. Where buildings are protected by an *automatic sprinkler system*, the outdoor storage, display and handling of combustible materials under eaves, canopies or other projections or overhangs are prohibited except where automatic sprinklers are installed under such eaves, canopies or other projections or overhangs.

C The storage or display of combustible materials beneath unsprinklered canopies or other building projections attached to an otherwise fully sprinklered building could lead to a rapidly developing fire in the stored material, which could gain sufficient headway beyond the capability of the building sprinkler system to suppress it, should it spread into the building’s interior. This section reinforces the requirements of NFPA 13 concerning the use of areas beneath building projections, such as eaves or canopies, for the storage or display of combustible materials where those locations are exempt from sprinkler protection as allowed in Section 8.15 of NFPA 13. NFPA 13 mandates that the scope of required sprinkler protection includes canopies or roofed-over areas attached to sprinklered buildings, unless these projections are constructed of noncombustible materials and the areas are not used for the storage, handling or display of combustible materials. Because NFPA 13 is a design standard and cannot be enforced as a maintenance document, this section essentially restates the NFPA 13 design requirement exception conditions in enforceable terms. Also note that, in the event that Appendix B of the code is adopted by a jurisdiction, areas used for the storage of combustible materials beneath a building’s horizontally projecting elements must be included in the building area for purposes of determining the required fire flow.

315.4.2 Height. Storage in the open shall not exceed 20 feet (6096 mm) in height.

C Storage pile height limitations are a means of controlling the size of potential fires and reducing tip-over potential, as well as a way to facilitate manual fire suppression by keeping the top of the pile within reach of conventional firefighting and overhaul tools, such as the ground ladders carried by an engine company or the long pike poles carried by ladder companies. The 20-foot (6096 mm) storage pile height limitation also correlates with Section 315.3 and helps reduce the likelihood that a fire would jeopardize adjacent properties in the event of a pile collapse.

315.5 Storage underneath high-voltage transmission lines. Storage located underneath high-voltage transmission lines shall be in accordance with Section 316.6.2.

C High-voltage transmission lines are a special hazard requiring specific regulations to prevent them from being exposed to fire conditions that could cause them to fail or fall to the ground. This general section directs the code user to the section containing more specific requirements for this topic.

315.6 Storage in plenums. Storage is prohibited in plenums. Abandoned material in plenums shall be deemed to be storage and shall be removed. Where located in plenums, the portion of abandoned cables that are able to be accessed without causing damage, or requiring demolition to the building shall be identified for future use with a tag or shall be deemed storage and shall be removed.

C This section introduces a concept that has long been in NFPA 70 and NFPA 90A—that plenums (see Chapter 2 commentary to the definition of “Plenum”) are intended specifically to be a part of the air distribution system. Plenums are also legitimately used for stringing communications and data cables as well as utility pipes, sprinkler pipes and similar items. However, such items often are not removed from the plenum when they become obsolete. For example, when an updated data

system is installed in a facility (which can occur every 18 to 24 months), it is not unusual for old wires to be cut out of the system but left in place with a new wiring system added on top of them (see Commentary Figure 315.6).

The suspended ceiling tile systems that often enclose plenums are not intended to support any significant weight and can, therefore, easily be overwhelmed by the added weight of storage or abandoned materials such as cables (see commentary, IBC Section 808.1.1.1). Recently, a plenum space fuel load study showed how the safety of firefighters is compromised by the weight of these abandoned cables. It pointed out that plenum space fuel loads and wiring issues are serious concerns for firefighters during interior firefighting operations and its key recommendation was that abandoned wiring be removed.

Although the primary reason to require the removal of abandoned materials in plenums is the elimination of unnecessary weight, fire safety considerations should also be taken into account, as should the practical consideration that the HVAC design airflow through the plenum could be adversely affected. Such a requirement was long believed not to be enforceable primarily because fire code inspectors would rarely spend their time looking into plenums in existing buildings. However, the danger presented by the fire load of storage or abandoned materials is well-documented. Thus, there should be no significant difficulty in having inspectors identify the existence of abandoned products—especially abandoned cables—classify them as storage and require their removal. Note that only the “accessible portions” of abandoned cables must be removed because there is no intent to cause potential damage to the building or facility by attempting to remove cables or circuits that are strung through walls, floors or other building elements. Where some cables are left in place because they are intended to be reused at a future date, they must be clearly identified as such by affixing a tag. See also Section 603.9.

Commentary Figure 315.6—Abandoned Data Cable in a Plenum



315.7 Outdoor pallet storage. Pallets stored outdoors shall comply with Sections 315.7 through 315.7.7. Pallets stored within a building shall be protected in accordance with Chapter 32.

C This section introduces the requirements for the outdoor storage of pallets. These requirements provide limitations and restrictions in order to address the high challenge fire protection issues involving large amounts of densely piled idle pallets. Note that this section is specific to pallets that are stored outdoors. The protection requirements for pallets that are stored indoors are provided in Chapter 32.

315.7.1 Storage beneath overhead projections from buildings. Where buildings are equipped throughout with an *automatic sprinkler system*, the outdoor storage of pallets under eaves, canopies or other projections or overhangs are prohibited except where automatic sprinklers are installed under such eaves, canopies or other projections or overhangs.

C This section makes it clear that if a building is protected with an indoor automatic sprinkler system, pallets are not allowed to be stored directly under eaves, canopies, projections or overhangs unless outside automatic sprinklers are provided under these construction elements.

315.7.2 Distance to lot line. Pallet storage shall not be located within 10 feet (3048 mm) of a *lot line*.

C This section requires that pallet storage areas be more than 10 feet from any lot lines in order to prevent fire spread to adjacent construction and to maintain sufficient separation distance from the hazard.

GENERAL REQUIREMENTS

315.7.3 Storage height. Pallet storage shall not exceed 20 feet (6096 mm) in height.

C This section limits the pallet storage height to be 20 feet or less order to limit the total storage volume within an outdoor storage area. Note that in addition to this height limit, Section 315.7.4 also limits the area of any individual pallet pile to 400 square feet or less.

315.7.4 Pallet pile stability and size. Pallet stacks shall be arranged to form stable piles. Individual pallet piles shall cover an area not greater than 400 square feet (37 m²).

C In addition to height limit of 20 feet or less provided in Section 315.7.3, this section limits the pallet storage area to be 400 square feet or less in order to limit the total storage volume within an individual outdoor storage pile area. In addition, this requirement allows for greater access to any stack within the storage area and minimizes the effect of tripping stacks on increasing the fire size.

315.7.5 Pallet types. Pallets shall be all wood, with slatted or solid top or bottom, with metal fasteners, or shall be plastic or composite pallets, *listed* and *labeled* in accordance with UL 2335 or FM 4996. Plastic pallets shall be both solid and gridded deck, independent of the pallet manufacturing process, type of resin used in fabrication or geometry of the pallet.

C This section limits the pallet types to be all wood, constructed as noted, or plastic or composite pallets, that are constructed as noted and are specifically listed and labeled in accordance with the referenced standards. These restrictions prohibit the outdoor storage of other, more hazardous types of pallets that have not been specifically fire tested in order to determine their performance and hazard level.

315.7.6 Pile separation distances. In addition to the other requirements of this section, pallet stacks and piles shall be separated in accordance with Sections 315.7.6.1 and 315.7.6.2.

C This section introduces the pile separation distance requirements that are listed in Tables 315.7.6(1) through 315.7.6(4).

TABLE 315.7.6(1)—SEPARATION DISTANCE BETWEEN WOOD PALLET STACKS AND BUILDINGS

WALL CONSTRUCTION	OPENING TYPE	WOOD PALLET SEPARATION DISTANCE (feet)		
		≤ 50 Pallets	51 to 200 Pallets	> 200 Pallets
Masonry	None	2	2	2
Masonry	Fire-rated glazing with open sprinklers	2	5	20
Masonry	Fire-rated glazing	5	10	20
Masonry	Plain glass with open sprinklers	5	10	20
Noncombustible	None	5	10	20
Wood with open sprinklers	—	5	10	20
Wood	None	15	30	90
Any	Plain glass	15	30	90

For SI: 1 foot = 304.8 mm.

TABLE 315.7.6(2)—SEPARATION DISTANCE BETWEEN PLASTIC PALLET STACKS AND BUILDINGS

WALL CONSTRUCTION	OPENING TYPE	PLASTIC PALLET SEPARATION DISTANCE (feet)		
		≤ 50 Pallets	51 to 200 Pallets	> 200 Pallets
Masonry	None	2	2	2
Masonry	Fire-rated glazing with open sprinklers	10	20	50
Masonry	Fire-rated glazing	15	40	100
Masonry	Plain glass with open sprinklers	15	40	100
Noncombustible	None	15	40	100
Wood with open sprinklers	—	15	40	100
Wood	None	30	80	150
Any	Plain glass	30	80	150

For SI: 1 foot = 304.8 mm.

TABLE 315.7.6(3)—SEPARATION FROM OTHER PALLET PILES AND ON-SITE STORAGE (WOOD PALLETS)

	WOOD PALLET SEPARATION DISTANCE (feet)		
	≤ 50 Pallets	51 to 200 Pallets	> 200 Pallets
Between pallet piles	7.5	15	45
Other on-site storage	7.5	15	45
For SI: 1 foot = 304.8 mm.			

TABLE 315.7.6(4)—SEPARATION FROM OTHER PALLET PILES AND ON-SITE STORAGE (PLASTIC PALLETS)

	PLASTIC PALLET SEPARATION DISTANCE (feet)		
	≤ 50 Pallets	51 to 200 Pallets	> 200 Pallets
Between pallet piles	15	40	75
Other on-site storage	15	40	75
For SI: 1 foot = 304.8 mm.			

315.7.6.1 Building separation. Pallet stacks and piles shall be separated from buildings in accordance with Table 315.7.6(1) for wood pallets and Table 315.7.6(2) for plastic pallets.

C This section provides direct reference to Tables 315.7.6(1) and 315.7.6(2). These tables list requirements based on building wall construction type, wall opening type and the number of total pallets that are stored. Table 315.7.6(1) is specific to wood pallets and Table 315.7.6(2) is specific to plastic pallets that meet the requirements of Section 315.7.5. These tables limit the pallet storage distances from any nearby building walls in order to prevent fire spread to adjacent construction on the same lot and to maintain sufficient separation distance from the hazard in order to maintain access.

315.7.6.2 Separation from other pallets and on-site storage. Pallets shall be separated from other pallet piles and other storage in accordance with Table 315.7.6(3) for wood pallets and Table 315.7.6(4) for plastic pallets.

C This section provides direct reference to Tables 315.7.6(2) and 315.7.6(4). These tables list requirements based on the distances between other pallet piles, as limited by Sections 315.7.3 and 315.7.4, other nonpallet storage areas and the number of total pallets that are stored in each individual pile. Table 315.7.6(3) is specific to wood pallets and Table 315.7.6(4) is specific to plastic pallets that meet the requirements of Section 315.7.5. These tables limit the pallet storage distances from other nearby pallet piles and nonpallet storage in order to prevent fire spread from any storage pile to an adjacent storage pile on the same lot and to maintain sufficient separation distance from the hazard in order to maintain access.

315.7.7 Prohibited locations. Pallets shall not be stored underneath high-voltage transmission lines, elevated roadways or elevated railways.

C This section prohibits the storage of pallets under the three listed critical transmission and transportation structures due to the increased potential hazardous spread of fire and disruption of service as well as the increased difficulty in accessing such a pallet storage area.

SECTION 316—HAZARDS TO FIREFIGHTERS

316.1 Trapdoors to be closed. Trapdoors and scuttle covers, other than those that are within a *dwelling unit* or automatically operated, shall be kept closed at all times except when in use.

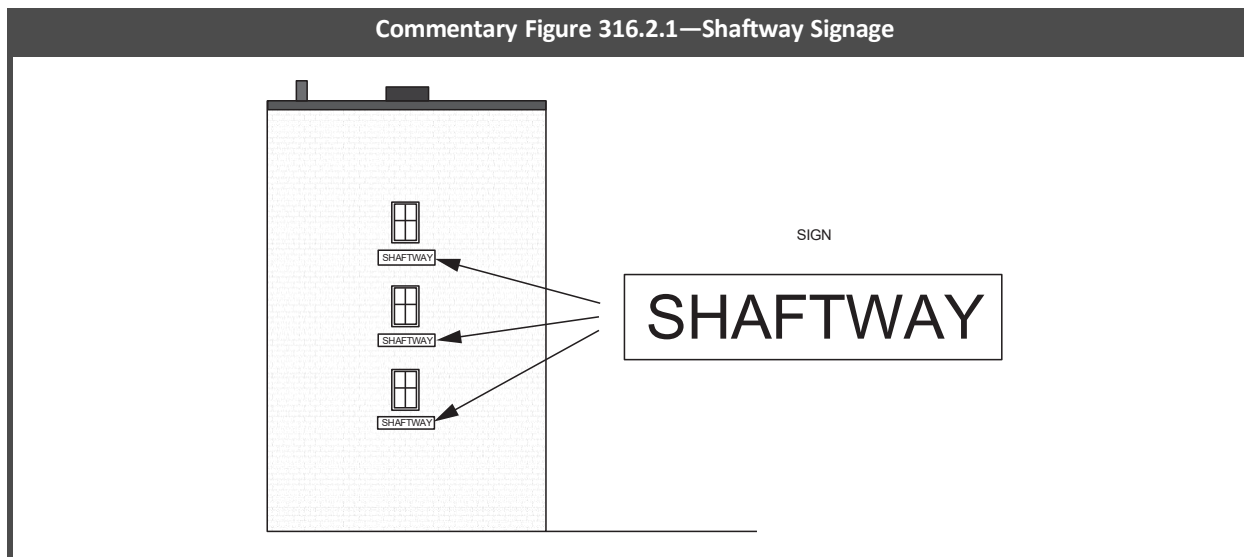
C Trapdoors and unguarded openings in floors and walkways must remain in the closed position or be designed to automatically close upon activation of the fire alarm system. Openings in floors or walkways can injure emergency responders, especially if vision is obscured.

316.2 Shaftway markings. Vertical shafts shall be identified as required by this section.

C This section was developed to prevent firefighters from falling through shafts when entering buildings off ladders placed on the exterior of the building.

316.2.1 Exterior access to shaftways. Outside openings that can be reached by the fire department and that open directly on a hoistway or shaftway communicating between two or more floors in a building shall be plainly marked with the word “SHAFTWAY” in red letters not less than 6 inches (152 mm) high on a white background. Such warning signs shall be placed so as to be readily discernible from the outside of the building.

C Exterior wall openings that are accessible to firefighters by way of ladders and aerial equipment and that open directly into shafts or hoistways communicating between two or more floors must be clearly marked (see Commentary Figure 316.2.1).



316.2.2 Interior access to shaftways. Door or window openings to a hoistway or shaftway from the interior of the building shall be plainly marked with the word “SHAFTWAY” in red letters not less than 6 inches (152 mm) high on a white background. Such warning signs shall be placed so as to be readily discernible.

Exception: Marking shall not be required on shaftway openings that are readily discernible as openings onto a shaftway by the construction or arrangement.

C Openings into shaftways from the interior of the building pose a threat to firefighters when visibility is poor. Interior shaft openings must be marked so that they are plainly visible from the interior of the building.

If firefighters can readily identify an opening into a shaft by the way the opening is constructed, the shaft opening need not be marked, keeping in mind that the firefighter may be feeling his or her way in heavy smoke or darkness.

316.3 Pitfalls. The intentional design or *alteration* of buildings to disable, injure, maim or kill intruders is prohibited. A person shall not install and use firearms, sharp or pointed objects, razor wire, *explosives*, *flammable* or *combustible liquid* containers, or dispensers containing highly toxic, toxic, irritant or other hazardous materials in a manner that could passively or actively disable, injure, maim or kill a firefighter who forcibly enters a building for the purpose of controlling or extinguishing a fire, rescuing trapped occupants or rendering other emergency assistance.

C This section prohibits the use of booby traps in buildings, for whatever reason, if they could injure or disable the emergency responder during the performance of their duties.

316.4 Obstructions on roofs. Wires, cables, ropes, antennas, or other suspended obstructions installed on the roof of a building having a roof slope of less than 30 degrees (0.52 rad) shall not create an obstruction that is less than 7 feet (2133 mm) high above the surface of the roof.

Exceptions:

1. Such obstruction shall be permitted where the wire, cable, rope, antenna or suspended obstruction is encased in a white, 2-inch (51 mm) minimum diameter plastic pipe or an *approved* equivalent.
2. Such obstruction shall be permitted where there is a solid obstruction below such that accidentally walking into the wire, cable, rope, antenna or suspended obstruction is not possible.

C Section 101.2 is intended, in part, to control conditions that can affect the safety of firefighters and emergency responders during operations. These conditions include obstructions on the roofs of buildings. This section does not prohibit the installation of these items, but it requires that they be identified or protected. The primary concern is poorly identified or unidentified obstructions on building roofs that can become an entanglement hazard or, in the case of equipment such as cellular or wireless communication antennas, a potential pathway for stray current if the electrical ground is somehow compromised and an emergency responder comes into contact with it. These provisions will also help to protect maintenance workers as they are working on a rooftop.

This section addresses obstructions on building roofs and was developed to establish criteria for identifying such obstructions. It is applicable only to roofs with a slope of 30 degrees (0.52 rad) or less. This section requires that a means of identification, a barrier, or some other form of obstruction be provided when a guy wire, cable, or rope is less than 7 feet (2133 mm) above the roof level. The 7-foot (2133 mm) value is based on the potential of a firefighter in a smoke-obscured environment striking the cable or wire, especially in the head, neck or upper torso area. A common fire service slang term for this is “being clotheslined” and these events have resulted in serious injury from entanglement or falling from a roof, including permanent paralysis of firefighters and other emergency responders.

Where the obstruction is located in the plane that is 7 feet (2133 mm) or less below the roof surface, Exception 1 allows the use of a protective collar such as a 2-inch (51 mm) diameter plastic pipe that is sleeved over the cable. Other options could include illuminating the cable, or providing reflective marking of the cable, wire or rope with a material similar to that used for marking of firefighter clothing and gear. Any method that offers equivalency to the prescribed pipe sleeve can be approved by the fire code official upon demonstration of equivalency.

Exception 2 recognizes that solid obstructions such as screen walls or fencing can provide a physical barrier so that walking under the wire is not possible and can eliminate the need for protecting the emergency responders from roof obstructions that are less than 7 feet (2133 mm) above the roof level.

316.5 Security device. Any security device or system that emits any medium that could obscure a *means of egress* in any building, structure or premise shall be prohibited.

C Security devices that, when activated, emit a medium such as smoke or other aerosols into a building could obscure exits or confuse occupants, thus creating an inherently dangerous situation for the public and responding emergency personnel. In cases of activation of these devices, armed criminal perpetrators could be trapped inside buildings. Law enforcement personnel arriving on the scene could easily believe that a building is on fire and responding firefighters could enter and be confronted by the perpetrator. Another danger is that false fire alarms could be transmitted automatically or by passers-by because of the appearance of smoke in the building. See also the commentary to Section 1032.2 regarding the reliability of exits.

316.6 Structures and outdoor storage underneath high-voltage transmission lines. Structures and outdoor storage underneath high-voltage transmission lines shall comply with Sections 316.6.1 and 316.6.2, respectively.

C This section introduces code provisions that address requirements for structure construction and material storage beneath high-voltage transmission lines (see definition in Section 202). The importance of such regulations and their impact on firefighter safety cannot be overstated. Among the various hazards firefighters face in the course of duty are electrical hazards during fire suppression or rescue activities. Performing fire-ground operations near power lines may expose firefighters to electric shock hazards through the following means, among others:

- Energized electrical lines or equipment obscured by dense smoke.
- Smoke and hot gases from a large fire creating a conductive path for electricity. When a fire is burning under a high-voltage transmission line, electricity could arc from the conductor to the ground, endangering people and objects near the arc.
- Making accidental contact with power lines that are still energized.
- Overhead power lines falling on and energizing conductive equipment and materials located on the fire ground.
- Electrical currents flowing through the ground and extending several feet (ground gradient).
- Solid-stream water applications on or around energized or downed power lines or equipment.

The regulations in Sections 316.6.1 and 316.6.2 are intended to minimize the noted exposures by prohibiting buildings, structures and storage from being constructed or maintained beneath high-voltage transmission lines except under the strictest of conditions and only with the fire code official's approval.

316.6.1 Structures. Structures shall not be constructed within the utility easement beneath high-voltage transmission lines.

Exception: Restrooms and unoccupied telecommunication structures of noncombustible construction less than 15 feet (4572 mm) in height.

C Structure fires underneath high-voltage transmission lines could cause arcing and shock hazard. Firefighting operations can involve the use of elevated platforms or aerial ladder apparatus and other emergency equipment, personnel above ground and hose streams that may come in close proximity to high-voltage transmission lines. According to nationally recognized utility companies, manual de-energization of lines may take 20 minutes or longer to accomplish.

The exception recognizes that certain ancillary structures, such as restroom buildings and telecommunications structures, do not generally present a significant exposure threat to overhead power lines.

316.6.2 Outdoor storage. Outdoor storage within the utility easement underneath high-voltage transmission lines shall be limited to noncombustible material. Storage of hazardous materials including, but not limited to, *flammable* and *combustible liquids* is prohibited.

Exception: Combustible storage, including vehicles and fuel storage for backup power equipment serving public utility equipment, is allowed, provided that a plan indicating the storage configuration is submitted and *approved*.

C This section regulates the fire load that can be allowed to exist beneath overhead high-voltage transmission lines by restricting outdoor storage within the easement to only noncombustible materials. The exception recognizes that, in certain instances, combustible materials may need to be stored within the easement, such as for utility work on the power lines. Such materials storage must be strictly regulated through a plans submittal and review process and approval by the fire code official. This section also makes it clear that under no circumstances are hazardous materials allowed to be stored beneath the power lines.

SECTION 317—VEGETATIVE AND LANDSCAPED ROOFS

317.1 General. Vegetative roofs and landscaped roofs shall comply with Sections 1505 and 1507.15 of the *International Building Code* and be installed and maintained in accordance with Sections 317.2 through 317.4.

C Among the major elements of green building construction are energy conservation and preservation of natural resources such as water. Rooftop vegetation accomplishes both using landscaping to cover part of a building roof. Vegetative and landscaped roofs offer a number of savings to the building owner. For buildings constructed with a membrane-type roofing system, landscaping can increase the life of the roofing materials by almost 100 percent according to some studies. Depending on the design, roof area and local climate, research has found certain green roofs can reduce summer cooling loads by about 25 percent and in some cases have reduced heat gain by 95 percent. In addition to reduced energy costs, these reductions have the potential to reduce the size of HVAC equipment, which in turn lowers the capital costs for construction. These roofs can also reduce the sound pressure level inside of the building and are helpful in controlling storm water runoff.

The requirements in this section limit the, require the use of roof assemblies designed for severe fire exposures and provide for the installation of a standpipe connection. It also sets forth requirements for establishing a maintenance plan for the vegetation installed on landscaped roofs. A landscaped roof is defined in Chapter 2. This section is intended to deal with the structural and fire safety concerns regarding construction that creates a green space on a building roof rather than the specific placement of any landscaping.

This section references the requirements in IBC Chapter 15 for landscaped roofs. IBC Section 1507.15 requires compliance with the requirements of the code, IBC Chapter 15 and specific structural requirements in IBC Chapter 16 for special purpose roofs and landscaped roofs. The IBC requires landscaped roofs to be calculated at a minimum 20 pounds per square foot (0.958 kN/m²) uniform live load and that the weight of landscaping materials be considered as dead loads and computed assuming the rooftop soil is saturated.

In addition to the IBC structural requirements, landscaped roofs can also require the installation of a listed Class A roof assembly. Section 317.1 of the code references the provisions in IBC Section 1505, which sets forth the fire exposure ratings for roof assemblies. The IBC fire classification requirements for roof assemblies are based on the building's construction type, not on the classification of the building's occupancy. The minimum roof covering classification prescribed by the IBC is either a Class B or Class C roof assembly. Class A roof assemblies are specified where the jurisdiction has adopted Appendix D of the IBC and the building is located in a fire district or where required in the IWUIC. IBC Section 1505.2 requires Class A assemblies to be listed and identified as such by an approved testing agency, but because the IBC does not normally require the installation of Class A roof assemblies, the number of listed systems available to design professionals will be limited. Additional requirements on the design of green roofs are contained in the *International Green Construction Code*® (IgCC®). Another source of information on "green roofs" is contained in Property Loss Prevention Data Sheet 1-35, Green Roof Systems, published by FM Global.

317.2 Vegetation. Vegetation shall be maintained in accordance with Sections 317.2.1 and 317.2.2.

C This section introduces requirements for the maintenance of rooftop gardens or landscaped roof areas to reduce the fire risk that could otherwise arise for buildings and exposure buildings because of the possible lack of maintenance. Basically, an unmaintained roof garden that catches fire could be likened to a brush fire being conducted on top of a building rather than on the ground, creating an increased exposure hazard since roof covering rating requirements are based on flying embers, not falling ones.

317.2.1 Irrigation. Supplemental irrigation shall be provided to maintain levels of hydration necessary to keep green roof plants alive and to keep dry foliage to a minimum.

C Care of rooftop vegetation by simple watering provides a basic line of defense against accumulation of dry biomass that can present a significant fire load if ignited. The frequency of watering is not specified due to the diverse moisture needs of the wide variety of planted materials possible on any given roof. The person responsible for the landscaped roof must determine the level of watering needed for this purpose and that information should be included in the maintenance plan authorized by Section 317.3.

317.2.2 Dead foliage. Excess biomass, such as overgrown vegetation, leaves and other dead and decaying material, shall be removed at regular intervals not less than two times per year.

C In order to manage the significant fuel load accumulation and fire hazard associated with dead or unkempt rooftop vegetation, this section sets a basic level of maintenance and regular removal of dead biomass, similar to that found in Section 304.1.2 for the maintenance of vegetation in and around a building (see commentary, Section 304.1.2).

317.3 Maintenance plan. The *fire code official* is authorized to require a maintenance plan for vegetation placed on roofs due to the size of a *vegetative roof* or *landscaped roof* area, materials used or where a fire hazard exists to the building or exposures due to the lack of maintenance.

C This section authorizes the fire code official to require a maintenance plan that may be used to guide maintenance personnel and inspectors in the care and upkeep of vegetation to reduce the hazard of having highly combustible dead plantings on the roof. This is beneficial to determine if the vegetative roof or landscaped roof will be maintained or is just a bunch of vegetation placed on the roof as a “do-it-yourself” project. The maintenance plan is also beneficial to determine what happens to a building that goes vacant or where the building owner does not keep up with maintenance.

317.4 Maintenance equipment. Fueled equipment stored on roofs and used for the care and maintenance of vegetation on roofs shall be stored in accordance with Section 313.

C This section simply sends the code user to Section 313 for appropriate storage regulations for fueled equipment (see commentary, Section 313).

SECTION 318—LAUNDRY CARTS

318.1 Laundry carts with a capacity of 1 cubic yard or more. Laundry carts with an individual capacity of 1 cubic yard [200 gallons (0.76 m³)] or more, used in laundries within Group B, E, F-1, I, M and R-1 occupancies, shall be constructed of noncombustible materials or materials having a peak rate of heat release not exceeding 300 kW/m² at a flux of 50 kW/m² where tested in a horizontal orientation in accordance with ASTM E1354.

Exceptions:

1. Laundry carts in areas protected by an *approved automatic sprinkler system* installed throughout in accordance with Section 903.3.1.1.
2. Laundry carts in coin-operated laundries.

C The change from cooking with animal fat to vegetable oil has increased the number of spontaneous ignition fires. These fires occur frequently in laundry operations due to insufficient cool-down time in the laundry drying process and have been known to ignite in freshly folded linen, often contained in polyethylene laundry carts. Medium-density polyethylene laundry carts which have a fuel content of approximately 20,020 Btu per pound [27 ft³ (0.76 m³)] are commonly used in industrial, commercial and institutional laundry operations. This material is essentially equal to the fuel value for gasoline and becomes a burning liquid that spreads and flows when involved in a fire. As such, the cart flammability hazard must be mitigated. This section does that by limiting combustible container heat release rate and heat flux using the test parameters of ASTM E1354. The peak rate of heat release criterion is consistent with the one in Section 304.3.2. The efficacy of using these lesser-hazard carts has been demonstrated in several commercial and institutional fire incidents wherein the burning commercial dryer contents were emptied into an ASTM E1354 container and taken outside and overhauled without evacuation of the building, loss to the structure or damage to the container.

Exception 1 permits the use and storage of ordinary laundry carts in areas protected by an approved sprinkler system. It would be up to the fire code official to determine the maximum number that could be stored under these conditions.

Exception 2 recognizes that the small carts used in neighborhood coin laundries do not present the same hazard as those used in commercial-type operations because the clothes and other laundered items do not remain in the cart for a significant length of time.

SECTION 319—ADDITIVE MANUFACTURING (3D PRINTING)

319.1 General. Additive manufacturing equipment and operations shall comply with Section 319.

C This section provides basic safety requirements for the use and operation of additive manufacturing, also referred to as 3D printing, in industrial and nonindustrial applications.

319.1.1 Scope. *Additive manufacturing* shall comply with one of the following:

1. Nonindustrial *additive manufacturing* shall comply with Section 319.2.
2. Industrial *additive manufacturing* shall comply with Section 319.3.

C This section provides the scope of the requirements for the two classifications of additive manufacturing. Nonindustrial additive manufacturing involves 3D-printing operations that do not create a hazardous (classified) location area outside the equipment and do not utilize an inert gas supply or a combustible dust collection system. Industrial additive manufacturing involves 3D-printing operations that utilize combustible powders or metals, an inert gas supply and/or a combustible dust collection system or that create a hazardous (classified) location area or zone outside the equipment.

GENERAL REQUIREMENTS

319.1.2 Installation, operation and maintenance. 3D printers and associated *additive manufacturing* equipment shall be installed, operated and maintained in accordance with this code, the listing and the manufacturer's instructions.

C This section requires that all production equipment to be used in the additive operation be specifically installed, operated and maintained such that it conforms to code requirements in addition to the listing requirements. The use of such equipment is to be limited to the way that it is specifically written in the manufacturer's instructions to ensure that the use of the equipment does not create an additional or unexpected hazard beyond what has been anticipated.

319.1.3 Production materials. Only the production materials *listed* for use with the equipment and included in the manufacturer's instructions shall be used.

C All production materials used in the additive operation must be specifically listed for use in the process. In addition, the use of such production materials is to be limited to those that are specifically listed in the manufacturer's instructions. The use of different materials that are not specifically listed could create an additional or unexpected hazard beyond what has been determined in the equipment listing.

319.2 Nonindustrial additive manufacturing. Nonindustrial *additive manufacturing* equipment and operations shall comply with Sections 319.2.1 and 319.2.2. *Additive manufacturing* equipment and operations that do not comply with Section 319.2 shall comply with Section 319.3.

C This section establishes basic safety requirements for self-contained equipment, which includes prepackaged production materials, in nonindustrial additive manufacturing. In addition, this section references Section 320.3 for industrial additive manufacturing.

319.2.1 Listing. 3D printers used in nonindustrial *additive manufacturing* shall be *listed* and *labeled* in accordance with UL 2011, UL 60950-1 or UL 62368-1. The listing shall also verify:

1. The 3D printers are self-contained and utilize maximum 30-liter prepackaged production materials.
2. The operation of the 3D printers shall not create a hazardous (classified) electrical area or zone outside the unit.
3. If any hazardous (classified) electrical area or zone exists inside the unit's outer enclosure, the area shall be protected by intrinsically safe electrical construction or other acceptable protection methods.
4. The 3D printers shall not utilize inert gas or an external *combustible dust* collection system.

C This section requires that four conditions must be verified for all 3D printers used in nonindustrial additive manufacturing. The 3D printers must also be listed and labeled in accordance with the UL standards stated. The product listing is relied on to verify that the equipment operates safely and does not create a hazardous (classified) area outside the unit.

319.2.2 Occupancies. Nonindustrial *additive manufacturing* shall be permitted in all occupancy groups.

C Entry-level 3D printers are commonly used in classrooms, offices and businesses for producing customized products and prototypes. This type of nonindustrial additive manufacturing typically occurs in low-production environments and does not create or use hazardous materials in or around the equipment. Based on the low hazard level, this type of additive manufacturing is allowed in all occupancy groups without requiring a change in the primary occupancy classification.

319.3 Industrial additive manufacturing. Industrial *additive manufacturing* equipment and operations shall comply with Sections 319.3.1 through 319.3.12.

C This section addresses industrial additive manufacturing operations, which include all operations that aren't covered by Section 319.2. These are typically industrial operations using external powder feed supplies, dust collection systems and/or inert gas supplies.

319.3.1 Permits required. Permits shall be obtained from the *fire code official* in accordance with Section 105.5 prior to engaging in industrial *additive manufacturing* operations.

C As with other operations that require specific review and approval based on the hazards present, an operational permit is required for all industrial additive manufacturing operations.

319.3.2 Listing. 3D printers used in industrial *additive manufacturing* shall be *listed* and *labeled* in accordance with UL 2011 or *approved* for the application based on a field evaluation conducted by an *approved* agency.

C This section requires industrial 3D printers to be listed and labeled but includes an option for nonlisted equipment to be approved based on a field evaluation conducted by an agency that is acceptable to the fire code official.

319.3.3 Combustible dusts and metals. Industrial *additive manufacturing* operations that store, use or produce *combustible dust*, combustible particulate solids or combustible metals shall comply with Chapter 22 and this section.

C Industrial additive manufacturing operations involving combustible dusts and metals must comply with the requirements of Chapter 22 in addition to the requirements of this section. See the commentary to Chapter 22 for more information.

319.3.4 Powder evaluation. Printing powders used in industrial *additive manufacturing* operations shall be tested for combustibility in accordance with NFPA 484 or NFPA 652 as applicable. A copy of test reports shall be provided to the *fire code official* upon request.

C Industrial additive manufacturing operations involving printing powders must be tested in accordance with NFPA 484 for combustible metals or NFPA 652 for combustible dust. As noted, the results of these tests are required to be provided to the fire code official. In addition, these test results are necessary in order to determine if other subsections of Section 319.3 require compliance.

319.3.5 Combustible (nonmetallic) dusts. Industrial *additive manufacturing* operations that store, use or produce combustible (nonmetallic) dusts shall comply with NFPA 654.

C This section requires compliance with NFPA 654 for the prevention of fire and dust explosions for industrial additive manufacturing operations involving combustible nonmetallic particulate dust solids.

319.3.6 Combustible metals. Industrial *additive manufacturing* operations that store or use combustible metals shall comply with NFPA 484.

C Industrial additive manufacturing operations specifically involving combustible metals must be protected from fires in accordance with NFPA 484.

319.3.7 Ancillary equipment. Ancillary equipment provided for recycling, sieving, vacuuming or handling combustible powders shall be designed and *approved* for such use.

C Any other equipment used as described in the industrial additive manufacturing operation must be specifically designed and intended for the way in which it is used in the process. In addition, the use of such equipment is to be reviewed and determined to be acceptable by the fire code official.

319.3.8 Hazardous materials. Industrial *additive manufacturing* operations that store or use hazardous materials exceeding the maximum allowable quantity limits shall comply with Chapter 50.

C Industrial additive manufacturing operations using hazardous materials exceeding the maximum allowable quantities must comply with the requirements of Chapter 50 in addition to the requirements of this section. See Chapter 50 and the commentary to Chapter 50 for more information.

319.3.9 Inert gas. Additive manufacturing processes that utilize inert gases shall comply with Chapter 53. Ventilation or gas detection shall be provided in accordance with Section 5307.

C Industrial additive manufacturing operations using inert gases must comply with the requirements of Chapter 53 in addition to the requirements of this section. See the commentary to Chapter 53 for inert gas requirements and Section 5307 for ventilation or gas detection requirements.

319.3.10 Technical assistance. Where required by the *fire code official*, a report evaluating the acceptability of technologies, processes, products, facilities, materials and uses associated with the operation shall be provided in accordance with Section 104.2.2 and *approved*.

C See the commentary to Section 104.2.2 for further explanation of the technical assistance report.

319.3.11 Performance-based design alternative. Where *approved* by the *fire code official*, buildings and facilities where industrial *additive manufacturing* is performed shall be permitted to comply with the performance-based design options in Section 5001.3 as an alternative to compliance with the other requirements set forth in this section.

C See the commentary to Section 5001.3 for more information regarding the performance-based design options allowed where approved by the fire code official.

319.3.12 Occupancies. Industrial *additive manufacturing* shall be conducted only in the occupancy groups associated with manufacturing operation and permitted by the Chapter 50 maximum allowable quantity tables. Where *approved*, the requirements in Section 319.3.6 shall be permitted to provide the technical basis for determining compliance with Table 5003.1.1(1), Note p.

C This section limits the industrial additive manufacturing operation to the same occupancy group as the overall manufacturing operation of the building and requires compliance with maximum allowable quantities provided in Chapter 50.

Note p of Table 5003.1.1(1) draws attention to the particularly challenging hazards of combustible dusts. This note specifies that where combustible dusts are manufactured, generated or used, and the concentration of dust or operating conditions create a fire or explosion hazard, the building or space must be classified as a Group H-2 occupancy. To avoid this occupancy classification, an analysis of the material's characteristics and the type of engineering and administrative controls that will be employed in the process to quantitatively mitigate the risk of a dust deflagration needs to demonstrate why such a classification is unwarranted. The basis for this determination must be documented in a technical report and opinion in accordance with Section 104.2.2 (or IBC Section 414.1.3). See also the commentary to the definition of "Combustible dust" in Chapter 2 and the commentary to Chapter 22, which specifically deals with dust.

SECTION 320—LITHIUM-ION AND LITHIUM METAL BATTERY STORAGE

320.1 General. The storage of lithium-ion and lithium metal batteries shall comply with Section 320.

Exceptions:

1. New or refurbished batteries installed in the equipment, devices or vehicles they are designed to power.
2. New or refurbished batteries packed for use with the equipment, devices or vehicles they are designed to power.
3. Batteries in original retail packaging that are rated at not more than 300 watt-hours for lithium-ion batteries or contain not more than 25 grams of lithium metal for lithium metal batteries.
4. Temporary storage of batteries or battery components during the battery manufacturing process prior to completion of final quality control checks.
5. Temporary storage of batteries during the vehicle manufacturing or repair process.

C Lithium-ion and lithium metal batteries can create challenging fire hazards. This section includes requirements that regulate the collection and storage of these batteries so as to reduce the probability of an event and mitigate any adverse impact on the affected facility and public safety. It is intended to cover all types of lithium-ion and lithium metal batteries (e.g., new, used, waste, refurbished), used batteries being collected for recycling or disposal, and batteries at recycling and disposal facilities. This section identifies the threshold quantities of batteries that are regulated by Section 320. As noted in the exceptions, it is not the intent to cover lithium-ion and lithium metal batteries in products, devices or vehicles, in small retail packaging, or the temporary storage of batteries at manufacturing facilities or in transit.

320.2 Permits. Permits shall be required for an accumulation of more than 15 cubic feet (0.42 m³) of lithium-ion and lithium metal batteries, other than batteries listed in the exceptions to Section 321.1, as set forth in Section 105.5.28.

C This section provides the quantity threshold for the amount of storage that requires a permit be issued. See also the commentary to Section 301.2.

320.3 Fire safety plan. A fire safety plan shall be provided in accordance with Section 404. In addition, the fire safety plan shall include emergency response actions to be taken upon detection of a fire or possible fire involving lithium-ion or lithium metal battery storage.

C This section provides the direct reference for the requirement that a fire safety plan be submitted as a part of the permit approval. See also the commentary to Section 403.10.6.

320.4 Storage requirements. Lithium-ion and lithium metal batteries shall be stored in accordance with Section 320.4.1, 320.4.2 or 320.4.3, as applicable.

C This section introduces the storage requirements for the indoor storage of lithium-ion and lithium metal batteries.

320.4.1 Limited indoor storage in containers. Not more than 15 cubic feet (0.42 m³) of lithium-ion or lithium metal batteries shall be permitted to be stored in containers in accordance with all of the following:

1. Containers shall be open top and constructed of noncombustible materials or shall be *approved* for battery collection.
2. Individual containers and groups of containers shall not exceed a capacity of 7.5 cubic feet (0.21 m³).
3. A second container or group of containers shall be separated by not less than 3 feet (914 mm) of open space or 10 feet (3048 mm) of space that contains combustible materials.
4. Containers shall be located not less than 5 feet (1524 mm) from *exits* or *exit access* doors.

C This section includes requirements for limited indoor battery storage in containers. This is primarily intended to cover the containers used to collect used batteries for recycling or disposal. These types of containers can be found in many establishments, including mercantile and other occupancies. There is a limitation for an individual container to not exceed 7.5 cubic feet, which is approximately the size of a 55-gallon barrel. A maximum 15 cubic feet of storage (two 55-gallon barrels) are allowed in this code application.

320.4.2 Indoor storage areas. Indoor storage areas for lithium-ion and lithium metal batteries, other than those complying with Section 320.4.1, shall comply with Sections 320.4.2.1 through 320.4.2.6.

C Indoor storage more than the limited amount in Section 320.4.1 is regulated in accordance with this section. It requires, among other things, an operational permit. These indoor storage areas, other than the limited storage applications covered by Section 321.4.1, are required to comply with all applicable requirements in Sections 321.4.2.1 through 321.4.2.6.

320.4.2.1 Technical opinion and report. A technical opinion and report complying with Section 104.2.2 shall be prepared to evaluate the fire and explosion risks associated with the indoor storage area and to make recommendations for fire and explosion protection. The report shall be submitted to the *fire code official* and shall require the *fire code official's* approval prior to issuance of a permit. In addition to the requirements of Section 104.2.2, the technical opinion and report shall specifically evaluate the following:

1. The potential for *deflagration* of flammable gases released during a thermal runaway event.

2. The basis of design for an *automatic sprinkler system* or other *approved* fire suppression system. Such design basis shall reference relevant full-scale fire testing or another *approved* method of demonstrating sufficiency of the recommended design.

C A technical opinion and report are required to be provided for the storage of lithium-ion and lithium metal batteries before a permit is issued. This is intended to provide the fire code official with a document that is specific to the intended storage configuration including all the details that are relevant to the design to provide a basis for approval. The opinion and report are to be prepared by parties that have the technical ability to evaluate the requirements listed and be acceptable to the fire code official.

320.4.2.2 Construction requirements. Where indoor storage areas for lithium-ion and lithium metal batteries are located in a building with other uses, battery storage areas shall be separated from the remainder of the building by 2-hour rated *fire barriers* or *horizontal assemblies*. *Fire barriers* shall be constructed in accordance with Section 707 of the *International Building Code*, and *horizontal assemblies* shall be constructed in accordance with Section 711 of the *International Building Code*.

Exceptions:

1. Where battery storage is contained in one or more *approved* prefabricated portable structures providing a complete 2-hour fire-resistance-rated enclosure, *fire barriers* and *horizontal assemblies* are not required.
2. Where battery storage is limited to new batteries in packaging that has been demonstrated to and *approved* by the *fire code official* as sufficient to isolate a fire in packaging to the package interior, *fire barriers* and *horizontal assemblies* are not required.

C This section provides the fire-resistance rating requirements if the indoor storage is in a building with other uses. See also the *IBC® Code and Commentary* for a discussion of Sections 707 and 711. The first exception allows the use of specifically approved and designed prefabricated portable structures in place of these requirements. These structures are recognized as providing the equivalent protection as the room construction requirements. The second exception recognizes the use of new battery types that have been designed with a specifically approved protective packaging in order to provide an alternative protection to the room construction requirements.

320.4.2.3 Fire protection systems. Indoor storage areas for lithium-ion and lithium metal batteries shall be protected by an *automatic sprinkler system* complying with Section 903.3.1.1 or an *approved* alternative fire suppression system. The system design shall be based on recommendations in the *approved* technical opinion and report required by Section 320.4.2.1.

C This section provides the direct reference to Section 903.3.1.1 for the automatic fire sprinkler system requirement. In place of this requirement, an alternative fire suppression system is allowed to be used if it has been approved. In addition, the system is to be in accordance with the recommendations in the approved technical opinion and report if it has further requirements.

320.4.2.4 Fire alarm systems. Indoor storage areas for lithium-ion and lithium metal batteries shall be provided with an *approved* automatic fire detection and alarm system complying with Section 907. The fire detection system shall use air-aspirating smoke detection, radiant energy-sensing fire detection or both.

C This section provides the direct reference to Section 907 for the automatic fire alarm and detection system requirements. In addition, the system is required to use one of the specific types of detection to provide an early system activation.

320.4.2.5 Explosion control. Where the *approved* technical opinion and report required by Section 320.4.2.1 recommends explosion control, explosion control complying with Section 911 shall be provided.

C This section provides the direct reference to Section 911 for the requirements if explosion control is recommended as stated. See also the commentary to Section 911.

320.4.2.6 Reduced requirements for storage of partially charged batteries. Indoor storage areas for lithium-ion and lithium metal batteries with a demonstrated state of charge not exceeding 30 percent shall not be required to comply with Sections 320.4.2.1, 320.4.2.2 and 320.4.2.5, provided that procedures for limiting and verifying that the state of charge will not exceed 30 percent have been *approved*.

C This section recognizes that lithium-ion batteries not exceeding 30 percent state of charge (SOC) have been shown to be less likely to undergo thermal runaway or propagate than fully charged batteries. The 30 percent SOC level is recognized by the U.S. Department of Transportation (DOT) and other transport agencies (e.g., Transport Canada, International Civil Aviation Organization) as providing an additional level of safety for shipping by air. This section allows the fire code official to waive specific protection requirements based on approval of the procedures used for limiting and verifying that the state of charge will not exceed 30 percent. It is assumed that the procedures used to verify the SOC can be periodically checked as part of the operational permit.

320.4.3 Outdoor storage. Outdoor storage of lithium-ion or lithium metal batteries shall comply with Sections 320.4.3.1 through 320.4.3.3.

C This section includes requirements for the outdoor storage of lithium-ion or lithium metal batteries. It addresses the following: location of storage in proximity to exposures, storage area size limits and separation, and fire detection requirements.

GENERAL REQUIREMENTS

320.4.3.1 Distance from storage to exposures. Outdoor storage of lithium-ion or lithium metal batteries, including storage beneath weather protection in accordance with Section 414.6.1 of the *International Building Code*, shall comply with one of the following:

1. Battery storage shall be located not less than 20 feet (6096 mm) from any building, *lot line*, public street, public alley, public way or *means of egress*.
2. Battery storage shall be located not less than 3 feet (914 mm) from any building, *lot line*, public street, public alley, public way or *means of egress*, where the battery storage is separated by a 2-hour fire-resistance-rated assembly without openings or penetrations and extending 5 feet (1524 mm) above and to the sides of the battery storage area.
3. Battery storage shall be located not less than 3 feet (914 mm) from any building, *lot line*, public street, public alley, public way or *means of egress*, where batteries are contained in *approved*, prefabricated portable structures providing a complete 2-hour fire-resistance-rated enclosure.

C This section provides the separation requirements for the outdoor storage of these battery types. This includes the direct reference to IBC Section 414.6.1 for weather protection. The intent is to prevent fires from spreading to nearby buildings and other areas. Item 1 mirrors the requirements in Section 5003.12 for hazardous material outdoor control areas. Item 2 allows for a reduction in the separation distance if a 2-hour fire-resistance-rated assembly is provided as stated. Item 3 is like Item 2, but instead of using a 2-hour fire-resistance-rated assembly, the batteries have to be inside of approved prefabricated portable structures with the same level of fire-rated resistance. This recognizes a specific type of battery enclosure that is required to be acceptable to the fire code official for use in place of the other options.

320.4.3.2 Storage area size limits and separation. Outdoor storage areas for lithium-ion or lithium metal batteries, including storage beneath weather protection in accordance with Section 414.6.1 of the *International Building Code*, shall not exceed 900 square feet (83.6 m²). The height of battery storage in such areas shall not exceed 10 feet (3048 mm). Multiple battery storage areas shall be separated from each other by not less than 10 feet (3048 mm) of open space.

C This section provides the outdoor storage limits for these types of batteries, including those that comply with IBC Section 414.6.1. These limits and separation requirements are based on height and area rather than battery capacity. Each area cannot exceed 900 square feet in area unless it is separated by 10 feet of open space to prevent fires from spreading between areas. In addition, the height of the storage cannot exceed 10 feet.

320.4.3.3 Fire detection. Outdoor storage areas for lithium-ion or lithium metal batteries, regardless of whether such areas are open, under weather protection or in a prefabricated portable structure, shall be provided with an *approved* automatic fire detection and alarm system complying with Section 907. The fire detection system shall use radiant energy-sensing fire detection.

C This section provides the direct reference to Section 907 for the automatic fire detection and alarm system requirements. In addition, because the storage is outdoors, the system is required to use radiant energy-sensing fire detection to provide system activation.

SECTION 321—ARTIFICIAL COMBUSTIBLE VEGETATION

321.1 Artificial combustible vegetation on roofs and near buildings. Artificial combustible vegetation exceeding 6 feet (1829 mm) in height and permanently installed outdoors within 5 feet (1524 mm) of a building or on the roof of a building shall comply with Section 807.4.1. The placement of artificial combustible vegetation shall also comply with Sections 806.3 and 807.4.2.

Exception: Artificial decorative vegetation located more than 30 feet (9144 mm) from the *exterior wall* of a building.

C This section references Chapter 8 for artificial combustible vegetation installations in close proximity to buildings or on the roofs of buildings, regardless of whether a building is occupied since the intent is structure protection and not occupant protection. The requirement does not apply to artificial vegetation that is less than 6 feet (1829 mm) in height. This is intended to be a reasonable height since it is representative of most common interior applications. The use of the terms “permanent” and “installed” inform the code user that transient items are not regulated by this section.

The minimum fire separation distance is required since exterior applications do not have the benefit of fire sprinkler protection. In addition, the section does not differentiate between types of construction or exterior finishes. The 5-foot distance is based on the required distance in IBC Table 705.8.

SECTION 322—POWERED MICROMOBILITY DEVICES

322.1 General. Lithium-ion and lithium metal battery *powered micromobility devices* shall be operated and maintained in accordance with this section.

Exceptions:

1. Storage, repair and charging in residential occupancies of *powered mobility devices*, provided that such devices are for personal use by its owner.
2. Charging of a single *powered mobility device* in any occupancy by its owner.

C This section provides requirements for the use of micromobility devices. There have been significant fires due to these devices, including a four-alarm fire in NYC. This section contains reasonable safety requirements for the indoor charging of

these devices with lithium-ion and lithium metal type batteries. The listing of the device and the associated charging equipment that is compatible with the device and its battery is also a critical safety requirement.

322.1.1 Prohibited locations. The use of a residential occupancy as a business for the charging of commercially owned *powered micromobility devices* as part of a rental or sales service shall not be permitted.

C This section prohibits the use of a residential occupancy for operating a rental, sales and service business for these devices based on the level of hazard and its impact on life safety. This is an occurrence that fire departments have reported.

322.2 Battery chargers and equipment. *Powered micromobility devices* shall be charged in accordance with their listing and the manufacturer's instructions using only the original equipment manufacturer-supplied charging equipment or charging equipment in accordance with the listing and manufacturer's instructions.

C The improper charging of these devices can lead to a thermal runaway incident. This section limits the charging to the specific conditions and equipment as stated to prevent this occurrence.

322.3 Listing. *Powered micromobility devices* shall be *listed* and *labeled* in accordance with UL 2272 or UL 2849, as applicable.

C This requirement prohibits the use of devices that are not listed and labeled in accordance with these standards. UL 2272 addresses the electrical drive train system including the battery system, other circuitry and electrical components for electric powered scooters, and other devices that are referred to as personal e-mobility devices as defined in the standard. UL 2849 addresses the electrical system of eBikes powered by a lithium-based, rechargeable battery. EBikes include both Electrically Power Assisted Cycle (EPAC – pedal assist) and non-pedal assist eBikes.

322.4 Battery charging areas. Where *approved*, *powered micromobility devices* shall be permitted to be charged in a room or area that complies with all of the following:

1. Only *listed* devices utilizing *listed* charging equipment shall be permitted to be charged.
2. Is provided with sufficient electrical receptacles to allow the charging equipment for each device to be directly connected to a receptacle. Extension cords and relocatable power taps shall not be used.
3. Storage of combustible materials, combustible waste or hazardous materials shall not be permitted.
4. The charging operation shall not be conducted in or obstruct any required means of egress.
5. Removable storage batteries shall not be stacked or charged in an enclosed cabinet unless the cabinet is specially designed and *approved* for such purpose.
6. A minimum distance of 18 inches (457.2 mm) shall be maintained between each removable storage battery during charging operations unless each battery is isolated from neighboring batteries by an *approved* fire-resistant material.
7. A minimum of 18 inches (457.2 mm) shall be maintained between the location of the battery on each *powered micromobility device* during charging operations.
8. The indoor room or area shall be protected by a *fire alarm system* utilizing air-aspirating smoke detectors or radiant energy-sensing fire detection.

C The charging areas for these devices are the most hazardous locations identified by this section based on the potential for a thermal runaway incident. This section includes detailed restrictions intended to minimize the likelihood of this occurrence. Item 1 is like the requirements in Sections 322.2 and 322.3. Item 2 is intended to prevent the overloading of any electric circuits or equipment beyond the rated use. Item 3 is a general housekeeping requirement that is like the general requirements in Section 304 to limit the potential for multiple hazards in the charging area. Item 4 is like the requirements in Section 1032.2. Item 5 is intended to prevent improper ventilation and charging in a configuration that is not in accordance with the manufacturers' requirements. Items 6 and 7 require a minimum separation distance of 18 inches to minimize the chances for adjacent batteries impacting each other during charging. Item 8 requires one of the specific types of fire alarm system detection to provide an early system activation.

322.5 Fire safety plan. A fire safety plan shall be provided in accordance with Section 403.10.6. In addition, the fire safety plan shall include emergency response actions to be taken upon detection of a fire or possible fire involving lithium-ion or lithium metal battery storage.

C A fire safety plan is required by this section based on the level of hazard and impact on life safety. The plan is to include the specific details for the types of batteries used as stated. See also the commentary to Section 403.10.6.

Bibliography

The following resource materials were used in the preparation of the commentary for this chapter of the code:

FACE-99-F47, *Six Career Fire Fighters Killed in Cold-storage and Warehouse Building Fire—Massachusetts*. Washington, DC: National Institute for Occupational Safety and Health, September 2000.

NFPA 550—12, *Fire Safety Concepts Tree*. Quincy, MA: National Fire Protection Association, 2012.

Property Loss Prevention Data Sheet 1-35, Green Roof Systems. Johnston, RI: FM Global, 2011.